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(54) **BLOCKCHAIN-BASED GENERATIVE MEDIA SYSTEM FOR REAL-WORLD ASSET TOKENIZATION**

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(63) Continuation of application No. 18/671,824, filed on May 22, 2024, now Pat. No. 12,361,048, which is a continuation of application No. PCT/US2023/066776, filed on May 9, 2023.

(60) Provisional application No. 63/364,931, filed on May 18, 2022.

**Publication Classification**

(51) **Int. Cl.**

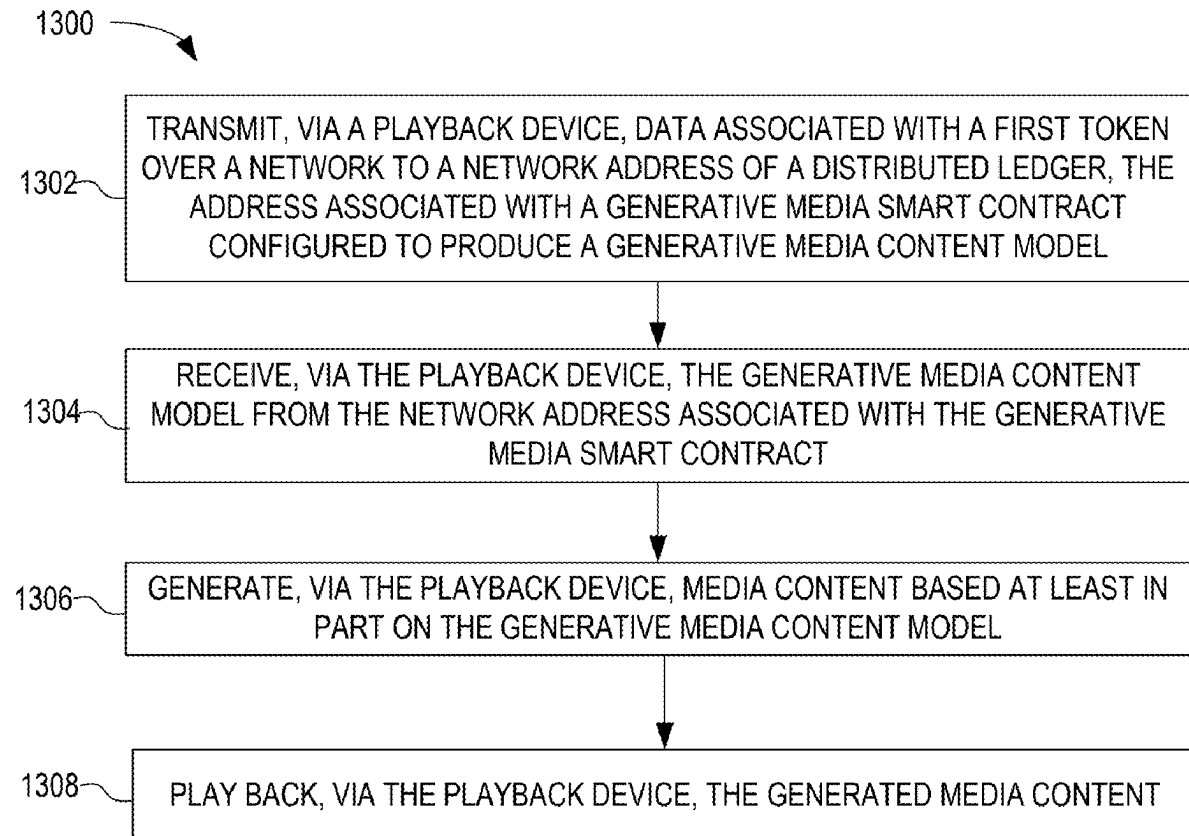
*G06F 16/41* (2019.01)

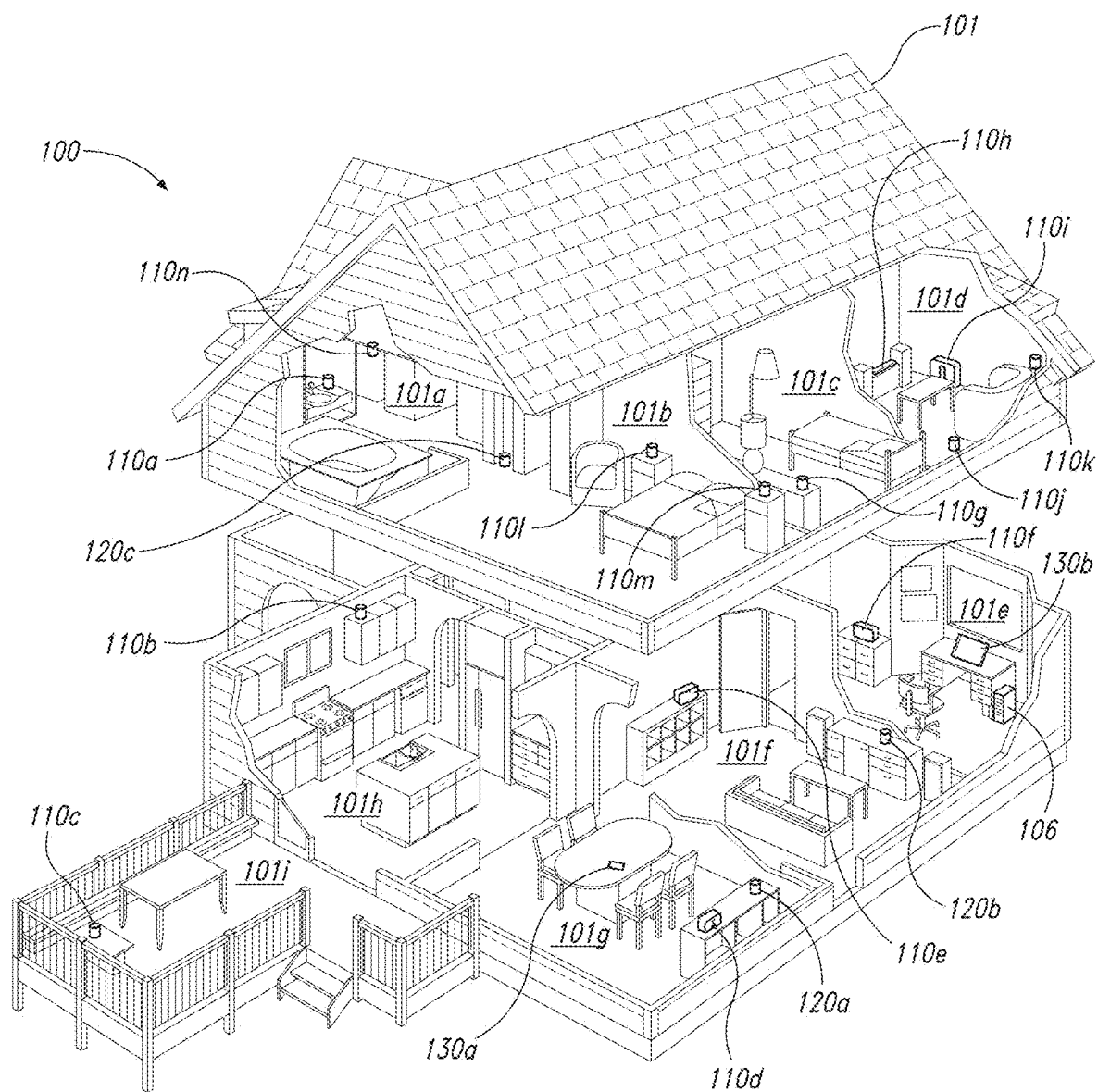
*G06F 3/16* (2006.01)

(57)

**ABSTRACT**

Generative media content (e.g., generative audio) can be dynamically generated based on various inputs, which can include blockchain data. A playback device accesses blockchain data stored via a distributed ledger and generates media content based at least in part on the blockchain data. The playback device can access a library of pre-existing media segments and arrange a selection of pre-existing media segments from the library for playback according to a generative media content model and based at least in part on the blockchain data. The generated media content can then be played back via the playback device.





*Fig. 1A*

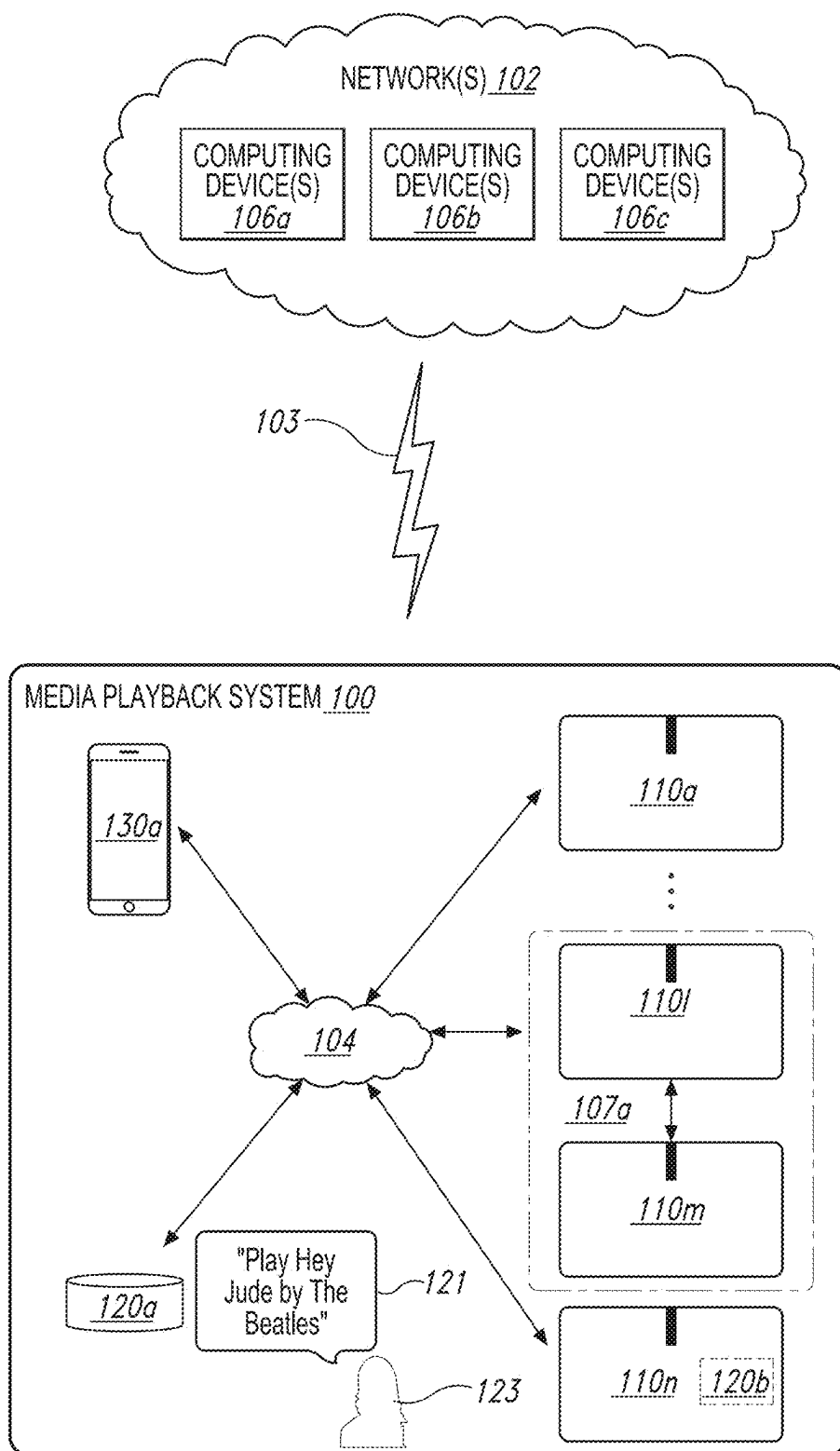


Fig. 1B

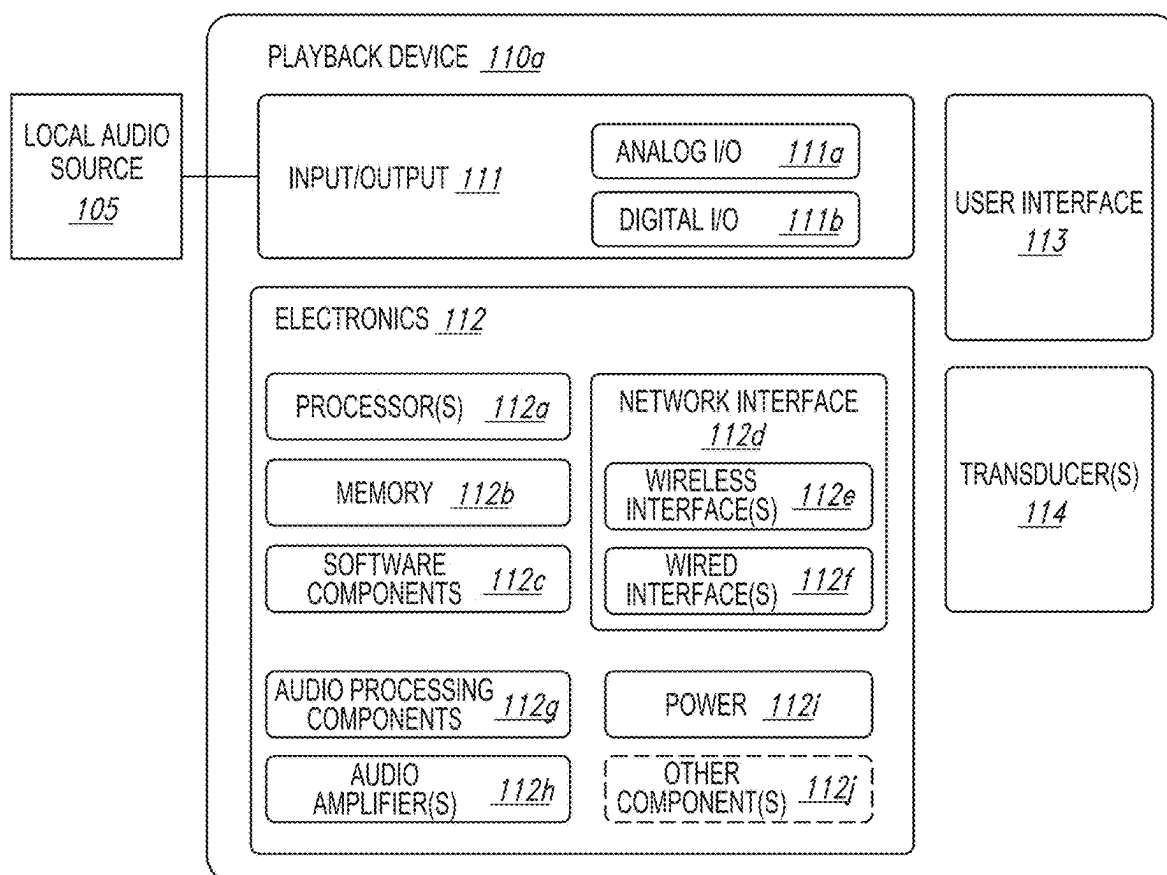


Fig. 1C

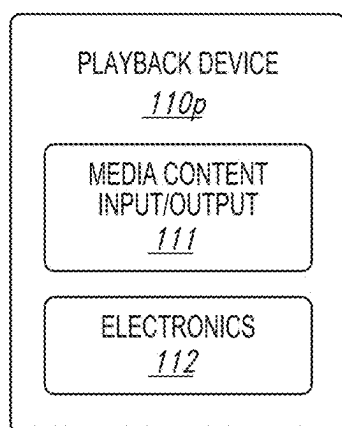


Fig. 1D

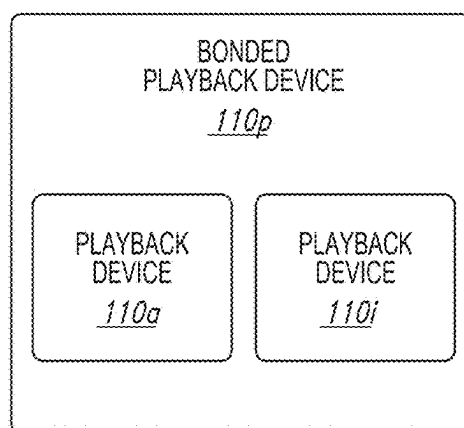
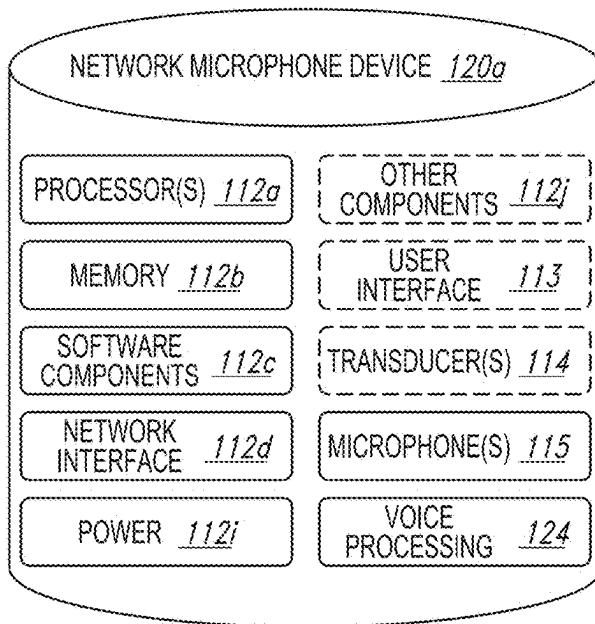
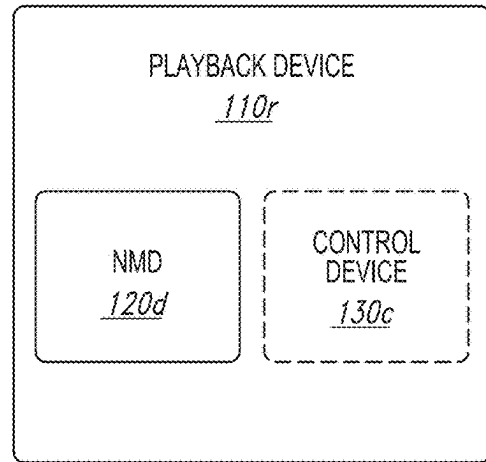


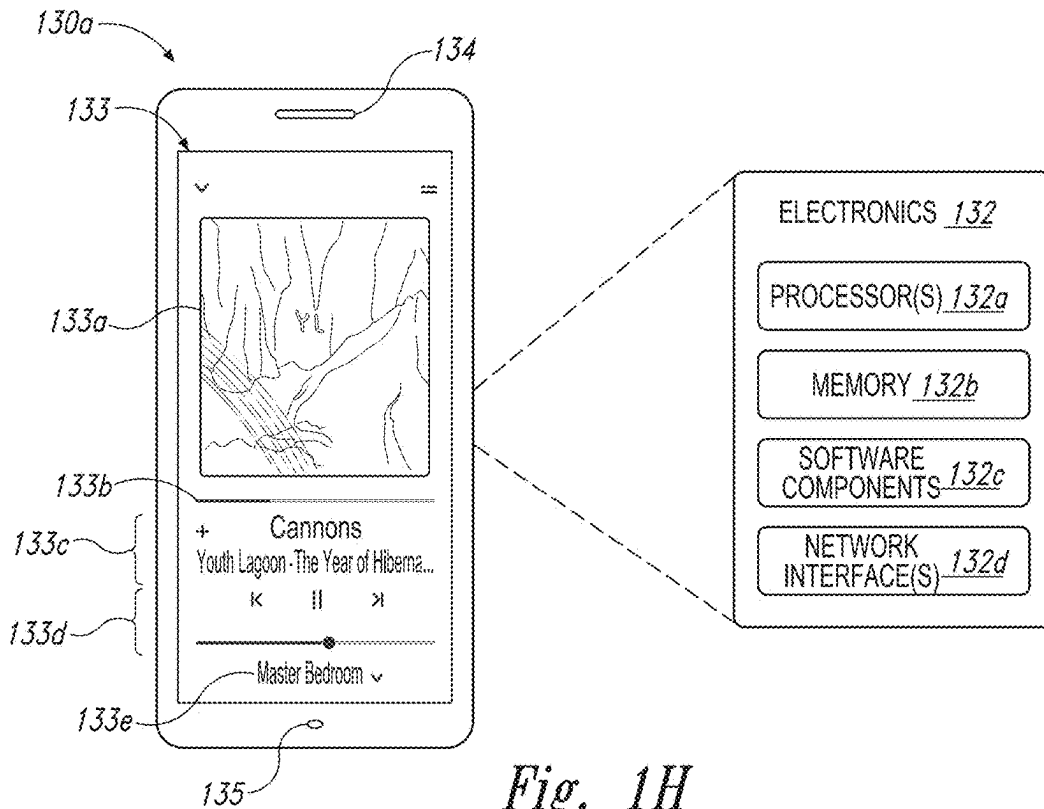
Fig. 1E



*Fig. 1F*



*Fig. 1G*



*Fig. 1H*



Fig. 1I

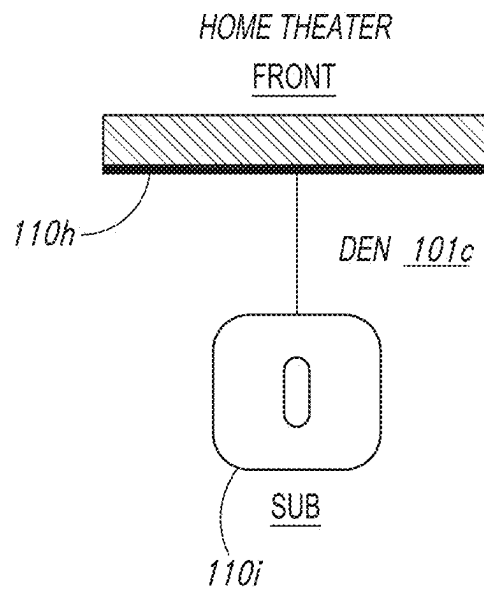


Fig. 1J

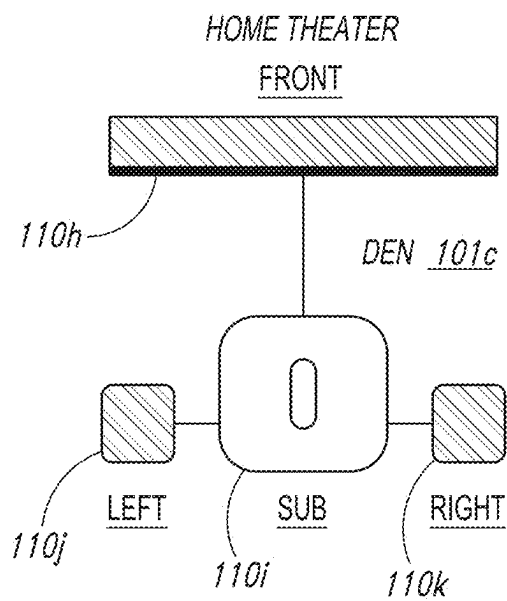


Fig. 1K

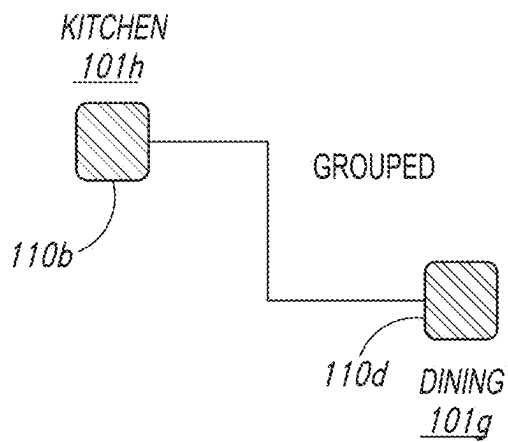


Fig. 1L

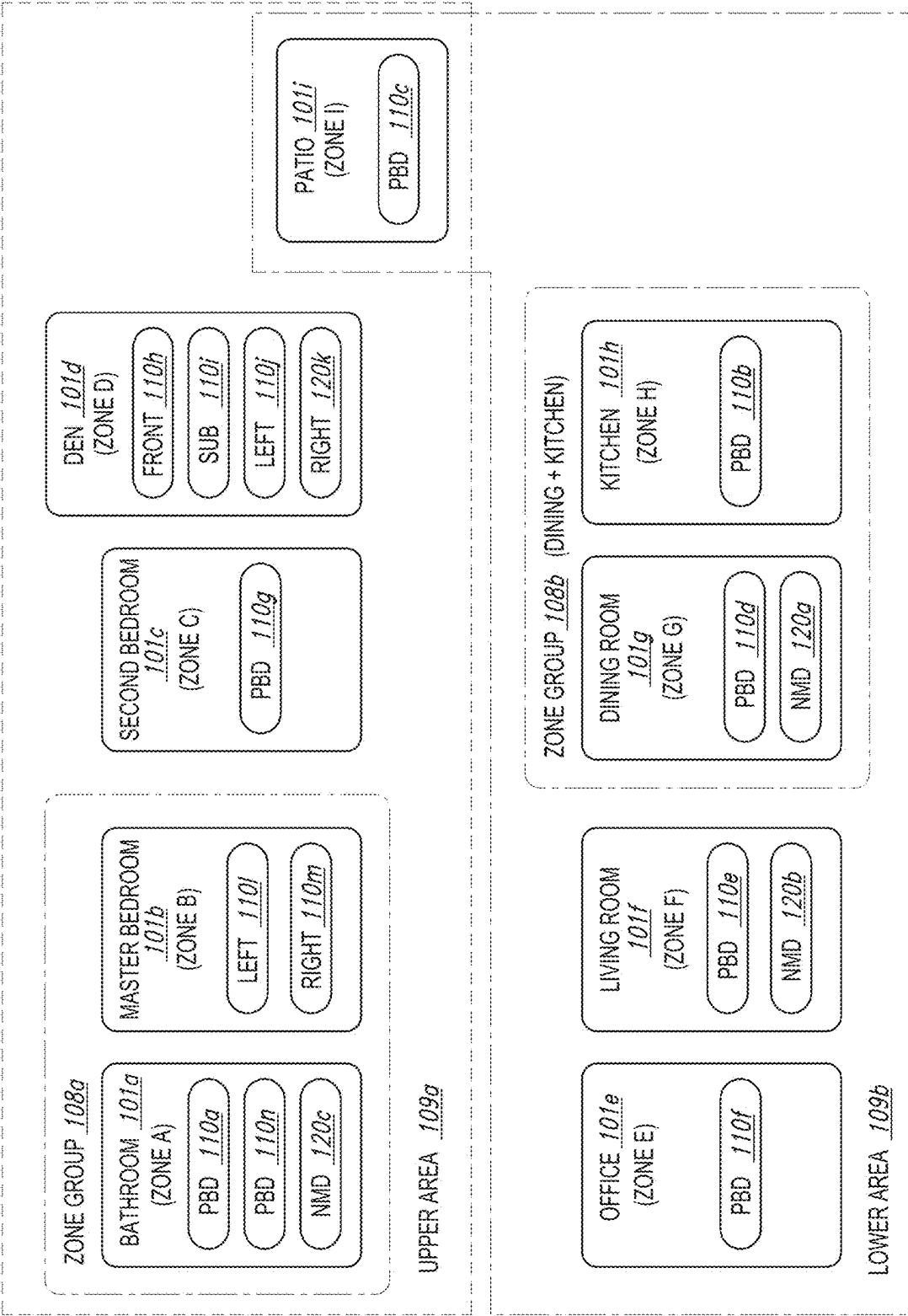


Fig. 1M

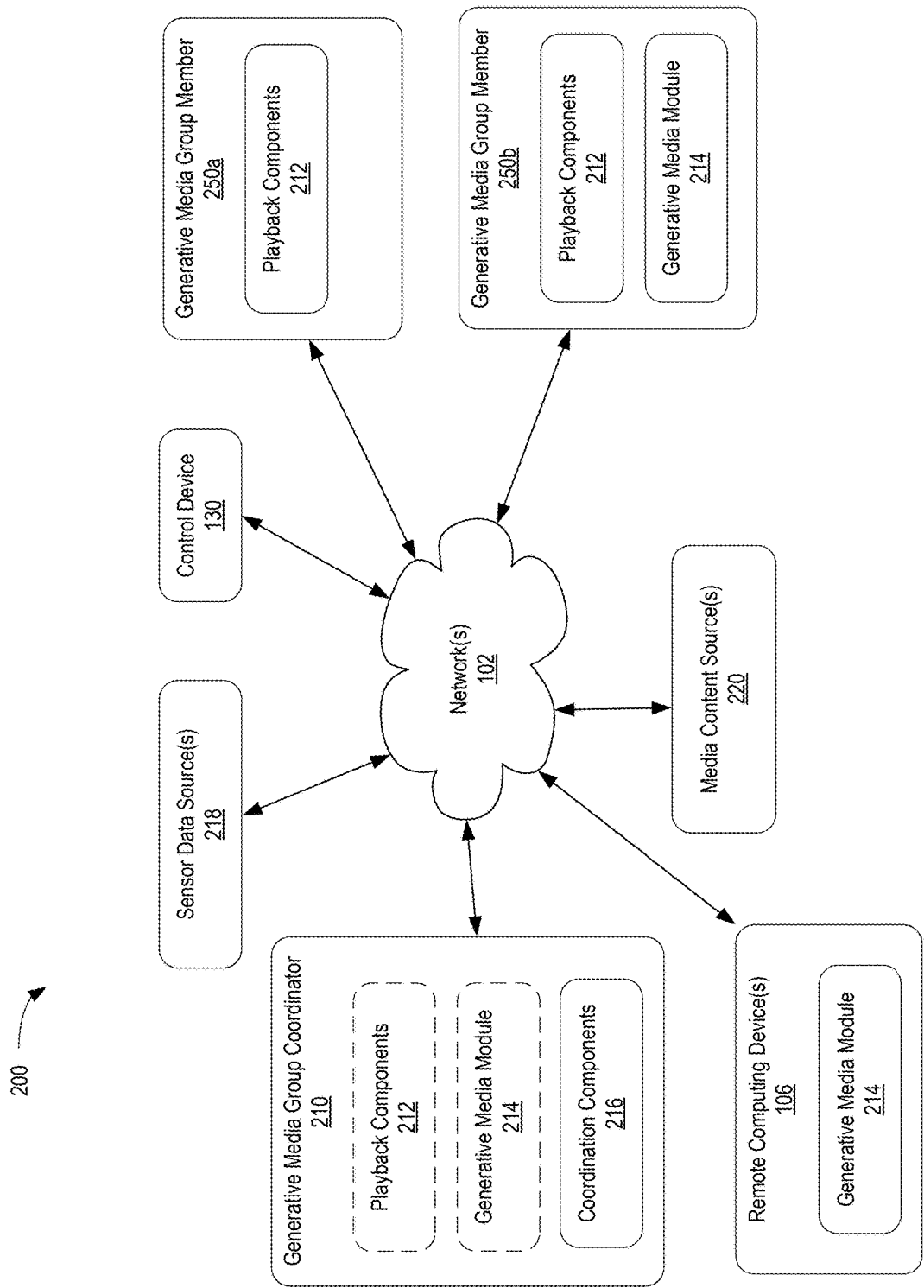


Fig. 2

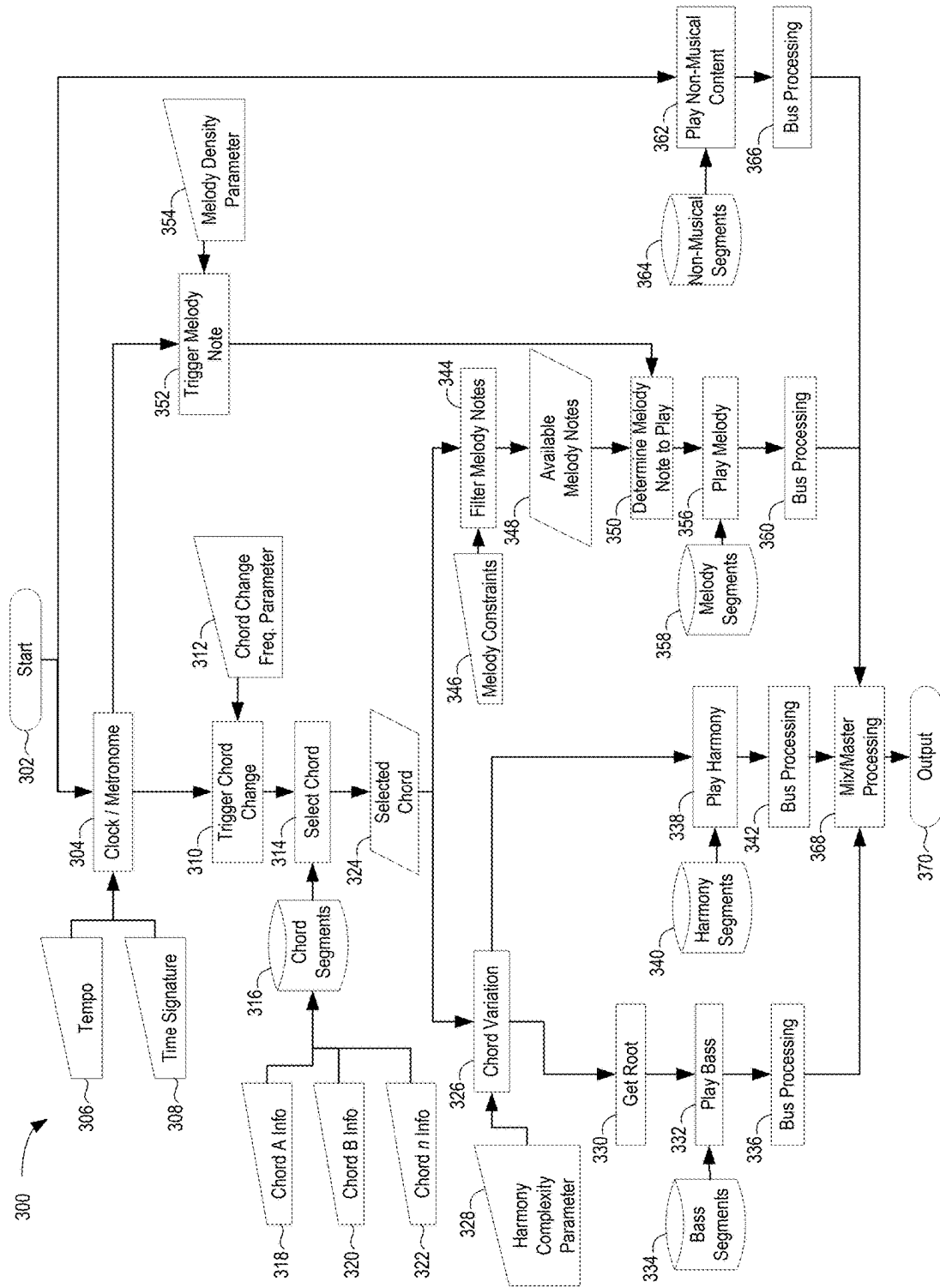


Fig. 3

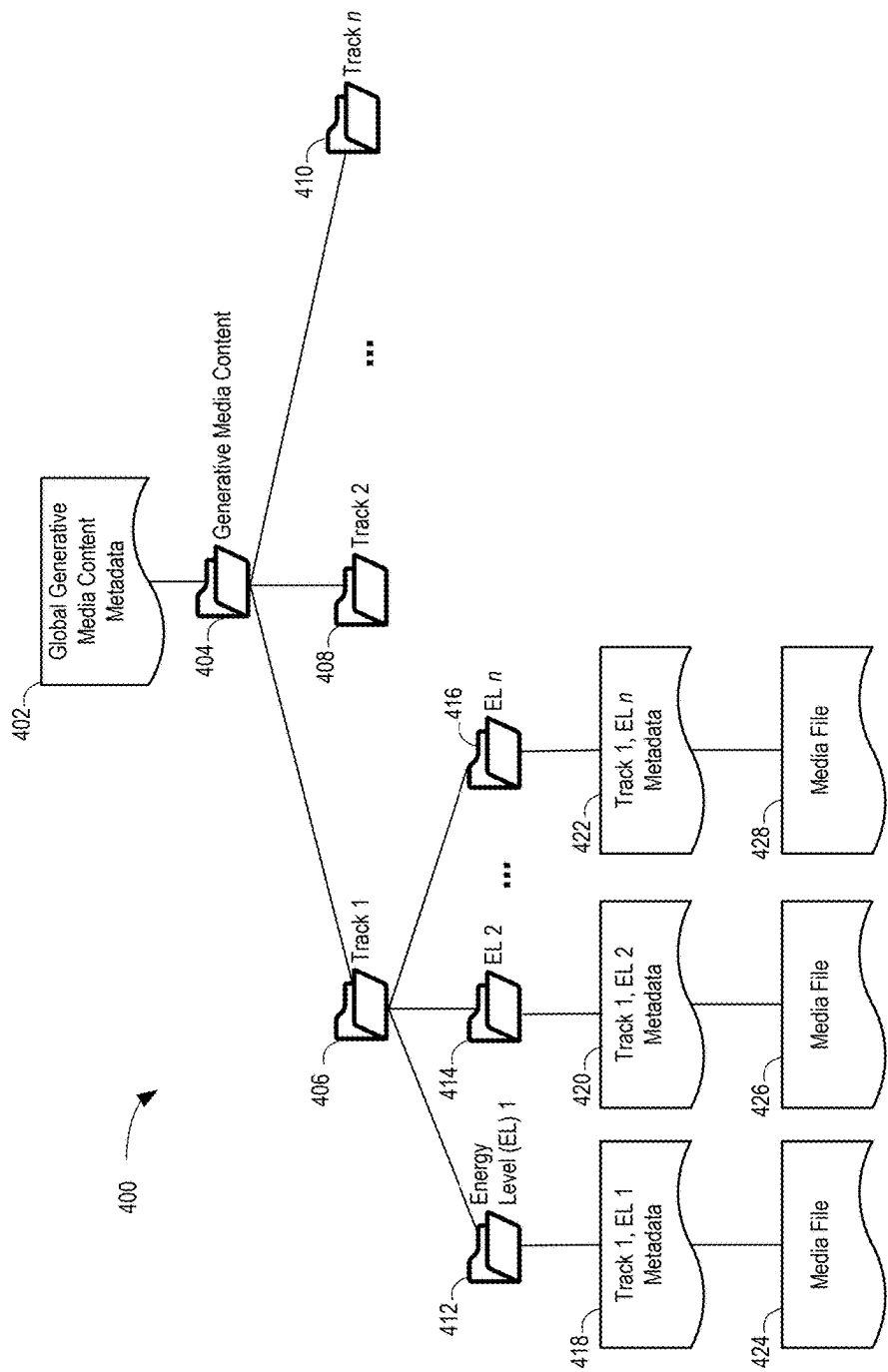
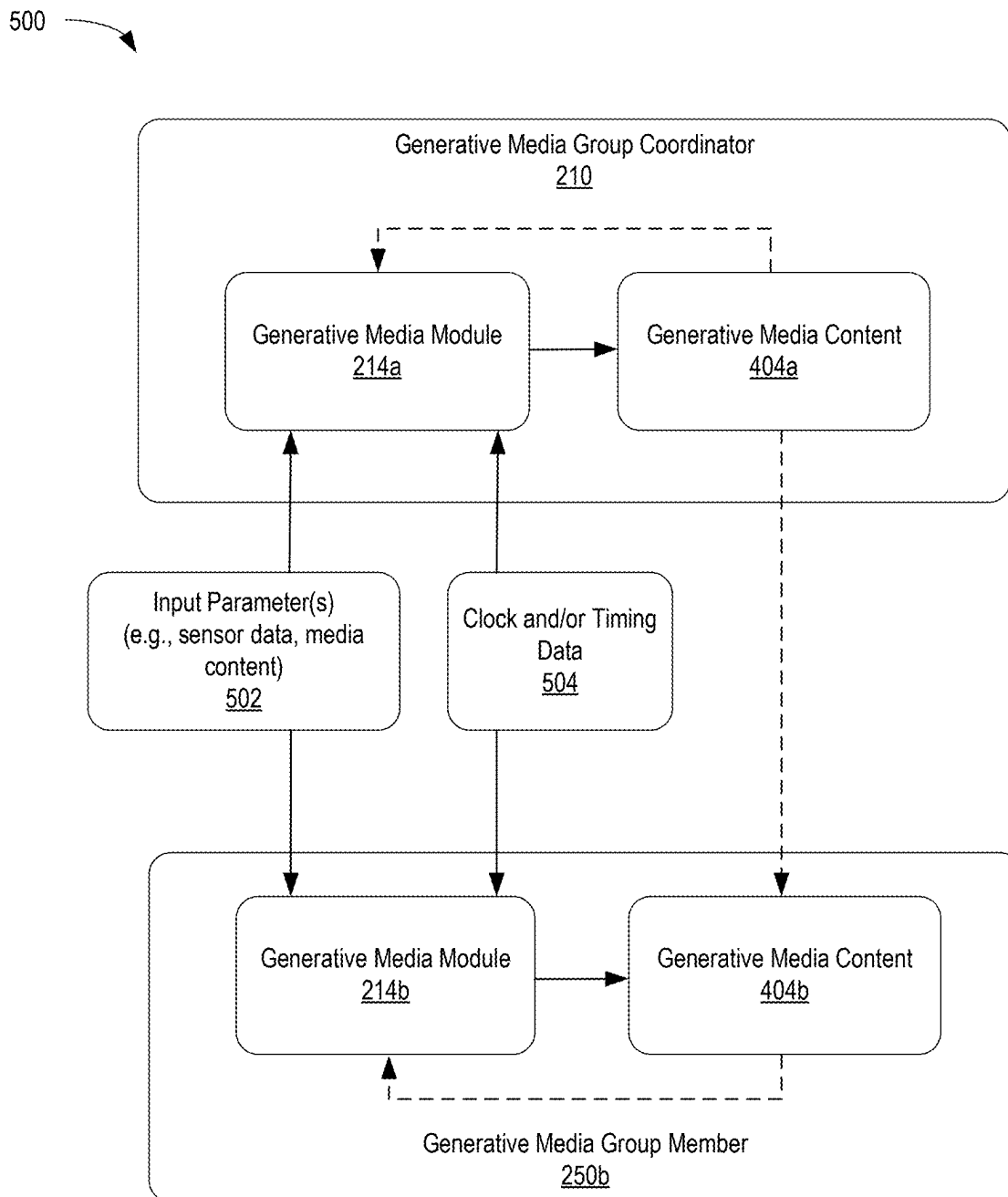


Fig. 4



*Fig. 5*

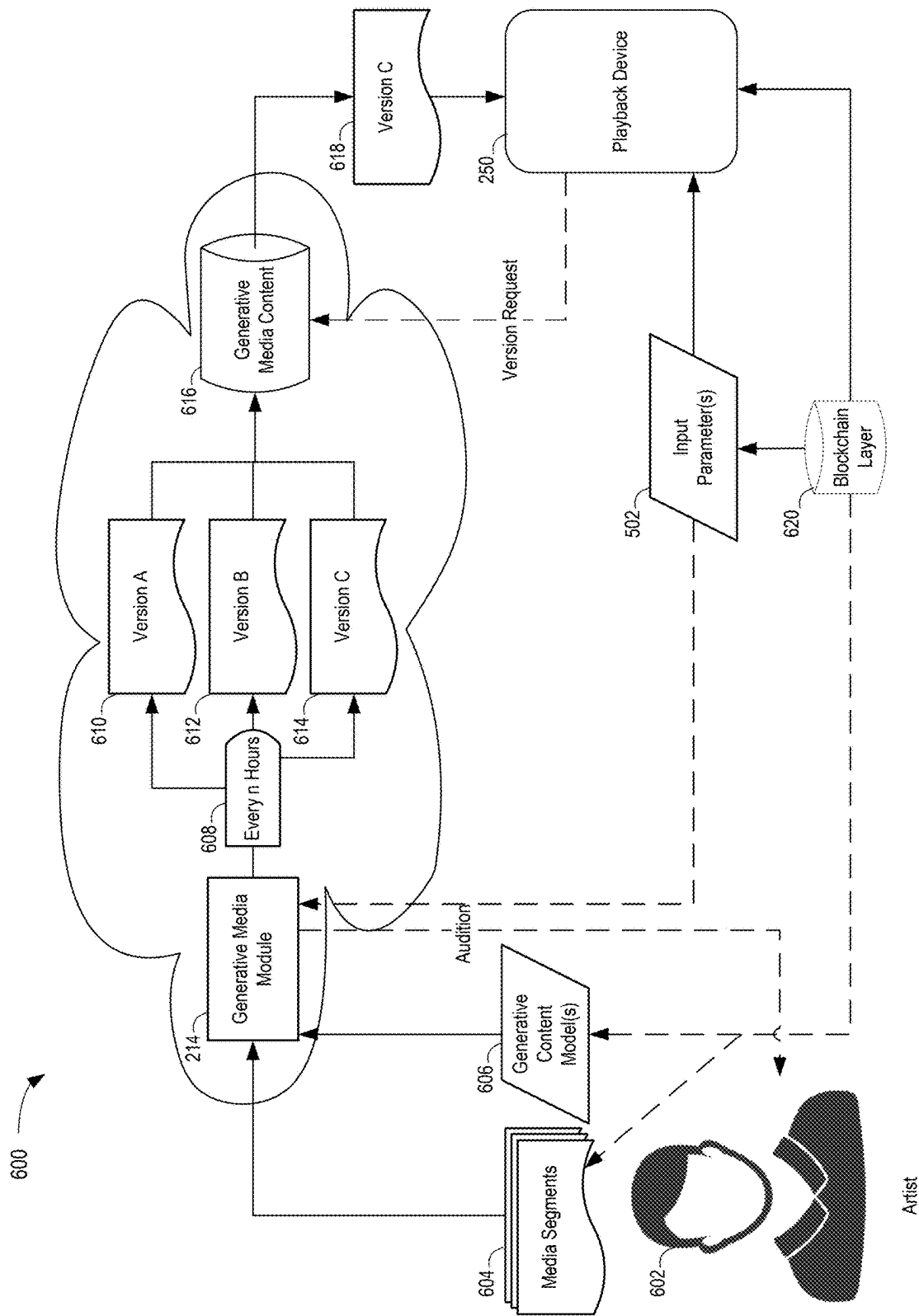


Fig. 6

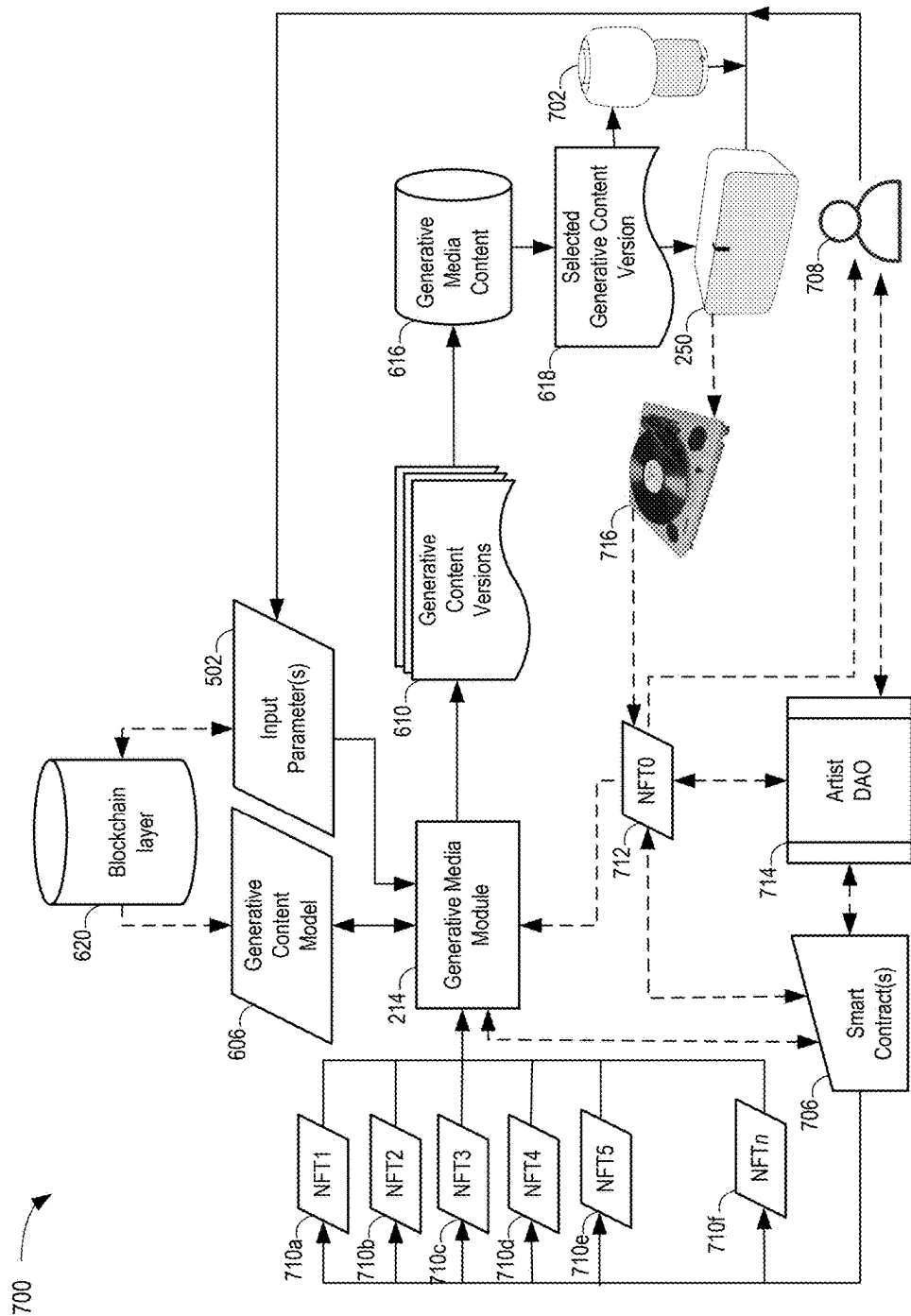
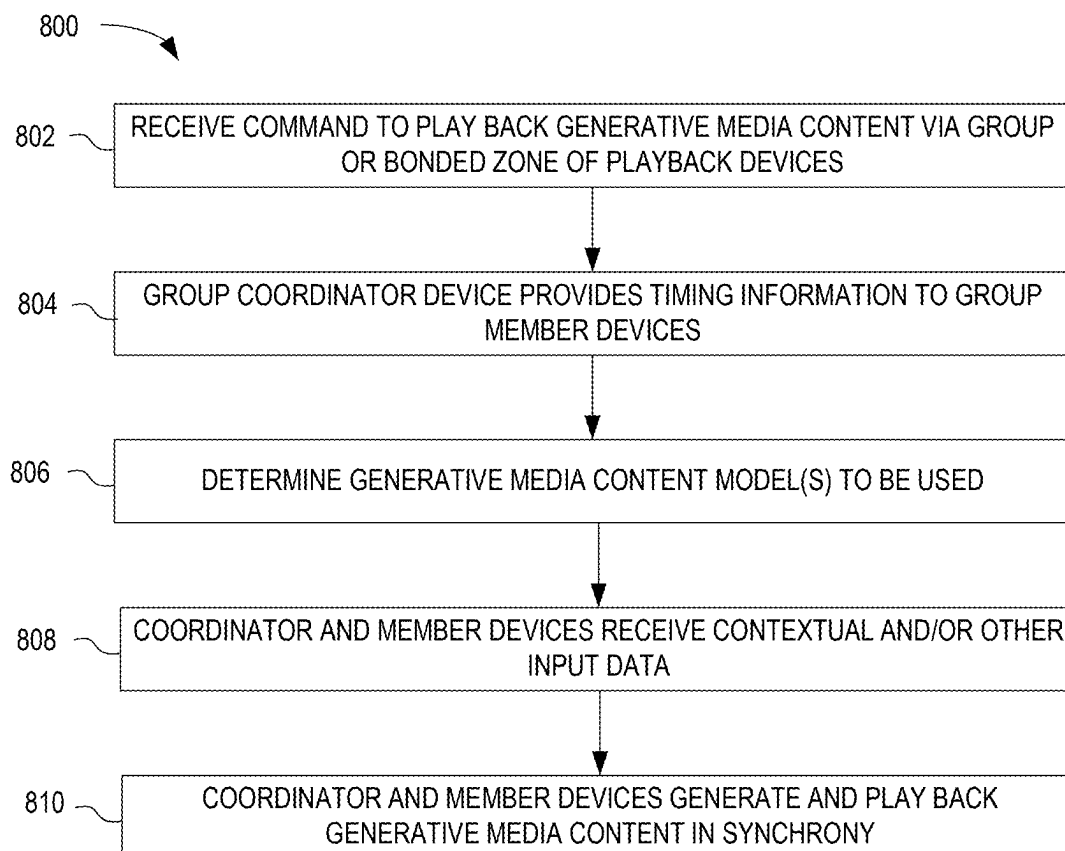
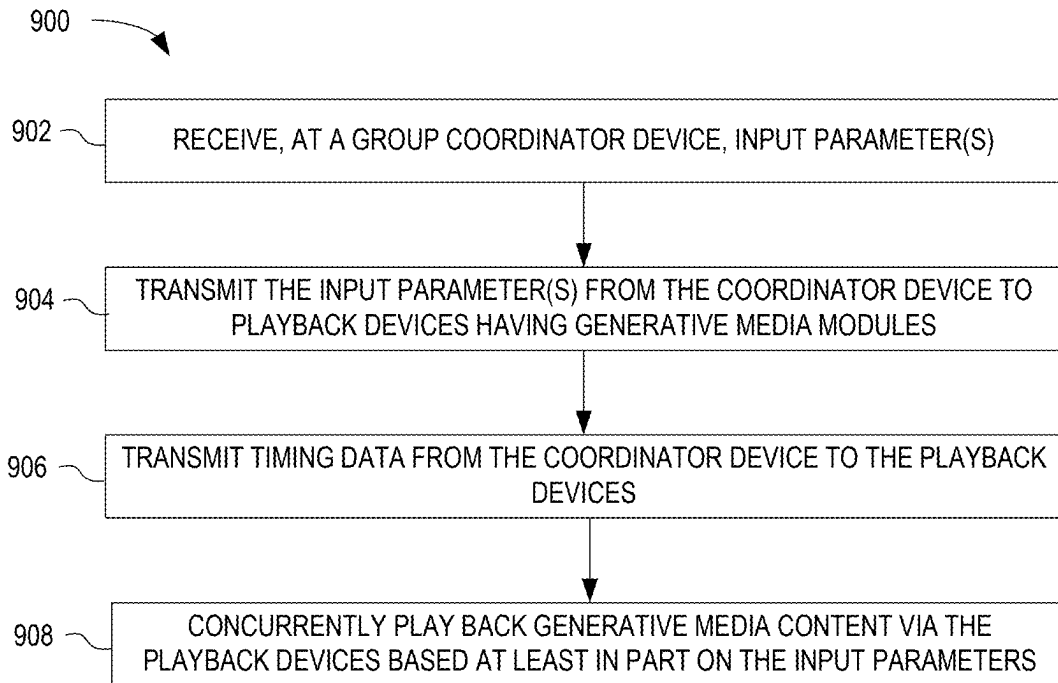
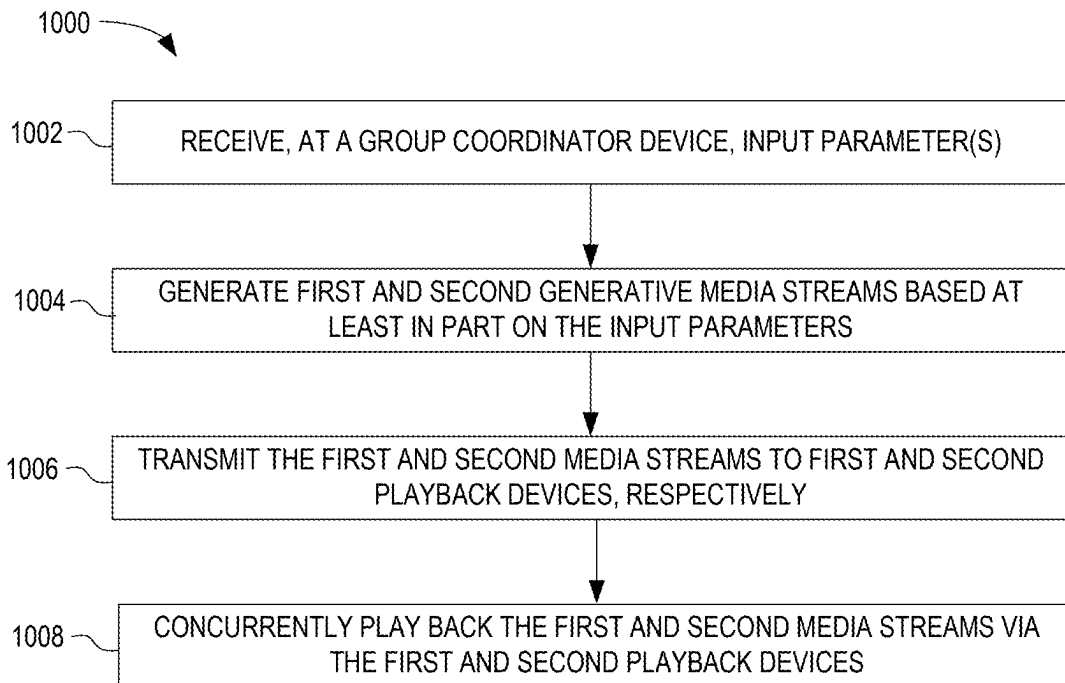
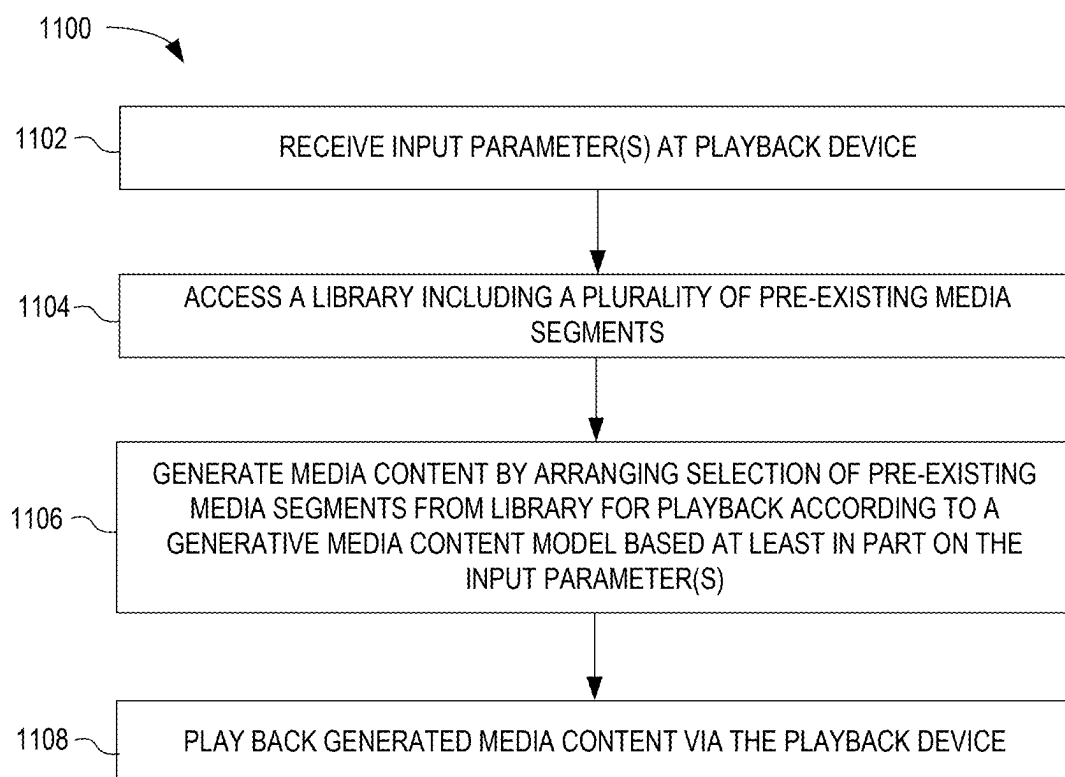


Fig. 7

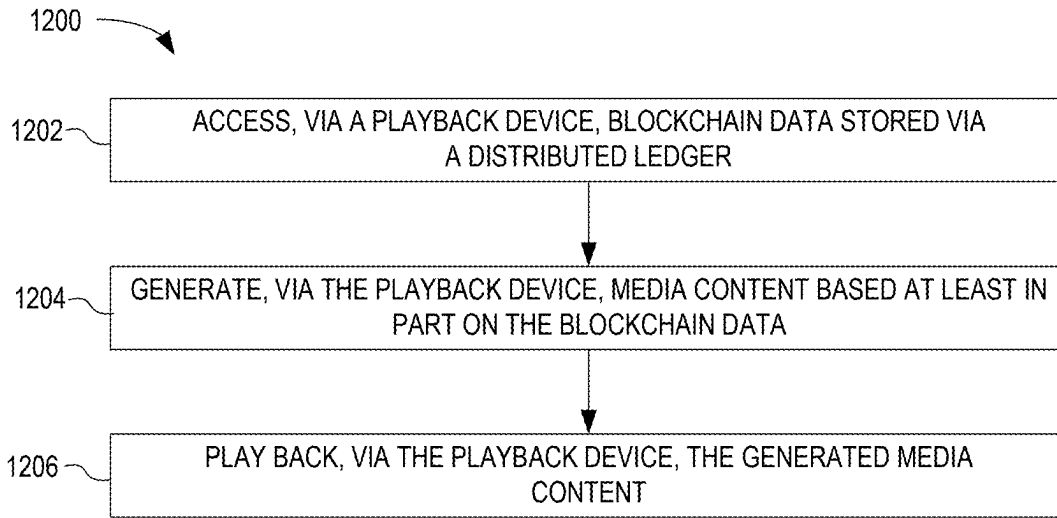
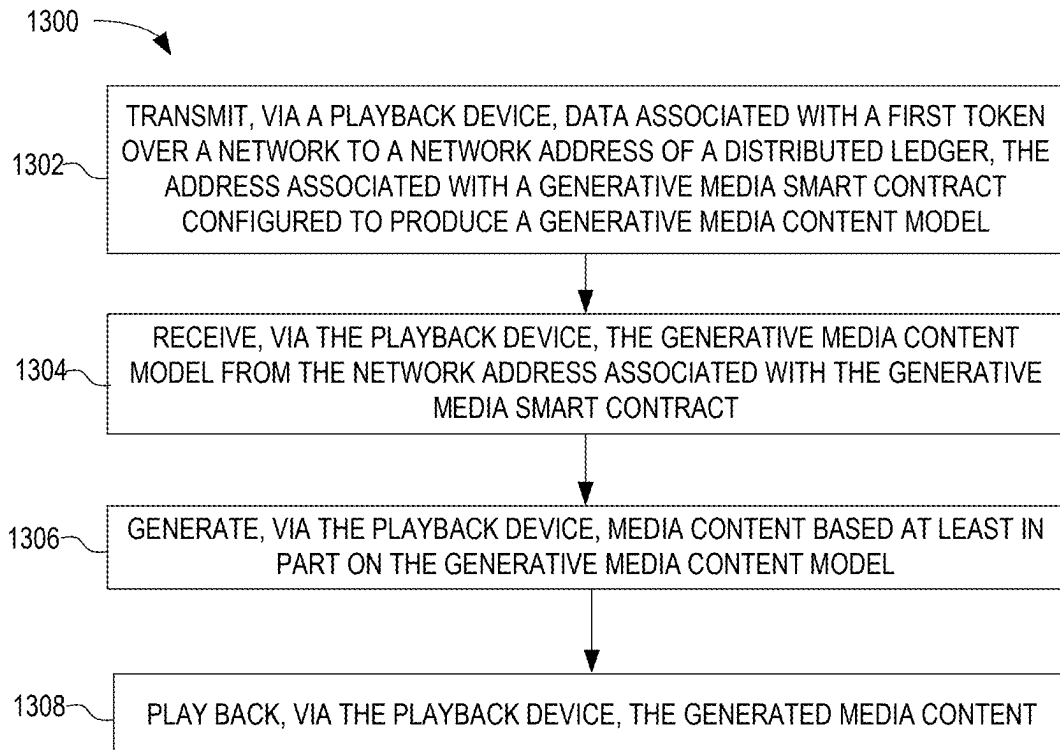


*Fig. 8*

*Fig. 9**Fig. 10*



*Fig. 11*

*Fig. 12**Fig. 13*

**BLOCKCHAIN-BASED GENERATIVE  
MEDIA SYSTEM FOR REAL-WORLD ASSET  
TOKENIZATION**

**CROSS-REFERENCE TO RELATED  
APPLICATIONS**

[0001] This application is a continuation of U.S. patent application Ser. No. 18/671,824, filed May 22, 2024, which is a continuation of International Patent Application No. PCT/US2023/066776, filed May 9, 2023, which claims the benefit of priority to U.S. Patent Application No. 63/364,931, filed May 18, 2022, each of which is incorporated herein by reference in its entirety.

**FIELD OF THE DISCLOSURE**

[0002] The present disclosure is related to consumer goods and, more particularly, to methods, systems, products, features, services, and other elements directed to media playback or some aspect thereof.

**BACKGROUND**

[0003] Options for accessing and listening to digital audio in an out-loud setting were limited until in 2002, when SONOS, Inc. began development of a new type of playback system. Sonos then filed one of its first patent applications in 2003, entitled “Method for Synchronizing Audio Playback between Multiple Networked Devices,” and began offering its first media playback systems for sale in 2005. The Sonos Wireless Home Sound System enables people to experience music from many sources via one or more networked playback devices. Through a software control application installed on a controller (e.g., smartphone, tablet, computer, voice input device), one can play what she wants in any room having a networked playback device. Media content (e.g., songs, podcasts, video sound) can be streamed to playback devices such that each room with a playback device can play back corresponding different media content. In addition, rooms can be grouped together for synchronous playback of the same media content, and/or the same media content can be heard in all rooms synchronously.

**BRIEF DESCRIPTION OF THE DRAWINGS**

[0004] Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following description, appended claims, and accompanying drawings, as listed below. A person skilled in the relevant art will understand that the features shown in the drawings are for purposes of illustrations, and variations, including different and/or additional features and arrangements thereof, are possible.

[0005] FIG. 1A is a partial cutaway view of an environment having a media playback system configured in accordance with aspects of the disclosed technology.

[0006] FIG. 1B is a schematic diagram of the media playback system of FIG. 1A and one or more networks.

[0007] FIG. 1C is a block diagram of a playback device.

[0008] FIG. 1D is a block diagram of a playback device.

[0009] FIG. 1E is a block diagram of a bonded playback device.

[0010] FIG. 1F is a block diagram of a network microphone device.

[0011] FIG. 1G is a block diagram of a playback device.

[0012] FIG. 1H is a partially schematic diagram of a control device.

[0013] FIGS. 1I through 1L show schematic diagrams of corresponding media playback system zones.

[0014] FIG. 1M shows a schematic diagram of media playback system areas.

[0015] FIG. 2 is a functional block diagram of a system for playback of generative media content in accordance with examples of the present technology.

[0016] FIG. 3 is a functional block diagram for a generative media module in accordance with aspects of the present technology.

[0017] FIG. 4 is an example architecture for storing and retrieving generative media content in accordance with aspects of the present technology.

[0018] FIG. 5 is a functional block diagram illustrating data exchange in a system for playback of generative media content in accordance with aspects of the present technology.

[0019] FIG. 6 is a schematic diagram of an example distributed generative media playback system in accordance with aspects of the present technology.

[0020] FIG. 7 is a schematic diagram of another example distributed generative media playback system in accordance with aspects of the present technology.

[0021] FIGS. 8-13 are flow diagrams of methods for playback of generative media content in accordance with aspects of the present technology.

[0022] The drawings are for the purpose of illustrating examples of the present technology, but those of ordinary skill in the art will understand that the technology disclosed herein is not limited to the arrangements and/or instrumentality shown in the drawings.

**DETAILED DESCRIPTION**

**I. Overview**

[0023] Generative media content is content that is dynamically synthesized, created, and/or modified based on an algorithm, whether implemented in software or a physical model. The generative media content can change over time based on the algorithm alone or in conjunction with contextual data (e.g., user sensor data, environmental sensor data, occurrence data). In various examples, such generative media content can include generative audio (e.g., music, ambient soundscapes, etc.), generative visual imagery (e.g., lighting, abstract visual designs that dynamically change shape, color, etc.), generative scent, generative tactile output (e.g., vibrations, haptic output, etc.), or any other suitable media content or combination thereof. As explained elsewhere herein, generative media can be created at least in part via an algorithm and/or non-human system that utilizes a rule-based calculation to produce novel media content.

[0024] Because generative media content can be dynamically modified in real-time, it enables unique user experiences that are not available using conventional media playback of pre-recorded content. For example, generative audio can be endless and/or dynamic audio that is varied as inputs (e.g., input parameters associated with user input, sensor data, media source data, or any other suitable input data) to the algorithm change. In some examples, generative audio can be used to direct a user's mood toward a desired emotional state, with one or more characteristics of the generative audio varying in response to real-time measure-

ments reflective of the user's emotional state. As used in examples of the present technology, the system can provide generative audio based on the current and/or desired emotional states of a user, based on a user's activity level, based on the number of users present within an environment, or any other suitable input parameter.

**[0025]** As another example, generative audio can be created and/or modified based on one or more inputs, such as a user's location, or activity, the number of users present in a room, time of day, or any other input (e.g., as determined by one or more sensors or by a user input). For example, when a single user is sitting at her desk in a calm state, the media playback system may automatically produce generative audio content suitable for focused study or work, whereas when multiple users are present in a room in an excited state with lots of movement, the same media playback system may automatically produce generative audio suitable for a social gathering or dance party. In various examples, audio characteristics that can be dynamically modified for producing generative audio can include selection of audio samples or clips, tempo, bass/treble/mid-range volume, spatial filtering of audio output, or any other suitable audio characteristics. The audio characteristics may be changed by using different tones or sounds, timing of the tones or sounds, and/or audio samples that may have the desired qualities. In some instances, characteristics may be changed by filtering or modulating playback of content as well, such as equalization, phase, or reverb/delay. During the listening experience, the audio characteristics of the generative music can be changed based on a number of inputs, such as time of day, geographic location, weather, or various user inputs, such as inferred mood, collective level of activity, or physiological inputs such as heart rate or the like.

**[0026]** In some instances, generative media content such as a soundscape can be associated with data stored in one or more blockchain databases and/or layers. A non-fungible token (NFT), for instance, generally comprises a data file stored on a blockchain that is associated with a digital or tangible asset (e.g., a song or album, visual artwork, literary work). While the NFT may refer to a digital work of art, for example, the NFT itself does not generally include the digital artwork itself since the file sizes required may be unwieldy (or too costly) to store on a blockchain layer. Instead, the NFT may comprise metadata (e.g., a URL or other locator) pointing to where the digital artwork is located and/or how to access the artwork. In various examples, an NFT or other data stored via a blockchain or other distributed ledger technology can be used for media content creation. For instance, in some examples an NFT serves as an input to a generative media engine, such that the resulting generative media content has properties that depend at least in part on the particular NFT or other blockchain data.

**[0027]** While some examples described herein may refer to functions performed by given actors such as "users," "listeners," and/or other entities, it should be understood that this is for purposes of explanation only. The claims should not be interpreted to require action by any such example actor unless explicitly required by the language of the claims themselves.

**[0028]** In the Figures, identical reference numbers identify generally similar, and/or identical, elements. To facilitate the discussion of any particular element, the most significant digit or digits of a reference number refers to the Figure in which that element is first introduced. For example, element

**110a** is first introduced and discussed with reference to FIG. 1A. Many of the details, dimensions, angles and other features shown in the Figures are merely illustrative of particular examples of the disclosed technology. Accordingly, other examples can have other details, dimensions, angles and features without departing from the spirit or scope of the disclosure. In addition, those of ordinary skill in the art will appreciate that further examples of the various disclosed technologies can be practiced without several of the details described below.

## II. Suitable Operating Environment

**[0029]** FIG. 1A is a partial cutaway view of a media playback system **100** distributed in an environment **101** (e.g., a house). The media playback system **100** comprises one or more playback devices **110** (identified individually as playback devices **110a-n**), one or more network microphone devices ("NMDs") **120** (identified individually as NMDs **120a-c**), and one or more control devices **130** (identified individually as control devices **130a** and **130b**).

**[0030]** As used herein the term "playback device" can generally refer to a network device configured to receive, process, and/or output data of a media playback system. For example, a playback device can be a network device that receives and processes audio content. In some examples, a playback device includes one or more transducers or speakers powered by one or more amplifiers. In other examples, however, a playback device includes one of (or neither of) the speaker and the amplifier. For instance, a playback device can comprise one or more amplifiers configured to drive one or more speakers external to the playback device via a corresponding wire or cable.

**[0031]** Moreover, as used herein the term NMD (i.e., a "network microphone device") can generally refer to a network device that is configured for audio detection. In some examples, an NMD is a stand-alone device configured primarily for audio detection. In other examples, an NMD is incorporated into a playback device (or vice versa).

**[0032]** The term "control device" can generally refer to a network device configured to perform functions relevant to facilitating user access, control, and/or configuration of the media playback system **100**.

**[0033]** Each of the playback devices **110** is configured to receive audio signals or data from one or more media sources (e.g., one or more remote servers or one or more local devices) and play back the received audio signals or data as sound. The one or more NMDs **120** are configured to receive spoken word commands, and the one or more control devices **130** are configured to receive user input. In response to the received spoken word commands and/or user input, the media playback system **100** can play back audio via one or more of the playback devices **110**. In certain examples, the playback devices **110** are configured to commence playback of media content in response to a trigger. For instance, one or more of the playback devices **110** can be configured to play back a morning playlist upon detection of an associated trigger condition (e.g., presence of a user in a kitchen, detection of a coffee machine operation). In some examples, for instance, the media playback system **100** is configured to play back audio from a first playback device (e.g., the playback device **110a**) in synchrony with a second playback device (e.g., the playback device **110b**). Interactions between the playback devices **110**, NMDs **120**, and/or control devices **130** of the media playback system **100**

configured in accordance with the various examples of the disclosure are described in greater detail below with respect to FIGS. 1B-1H.

**[0034]** In the illustrated example of FIG. 1A, the environment **101** comprises a household having several rooms, spaces, and/or playback zones, including (clockwise from upper left) a master bathroom **101a**, a master bedroom **101b**, a second bedroom **101c**, a family room or den **101d**, an office **101e**, a living room **101f**, a dining room **101g**, a kitchen **101h**, and an outdoor patio **101i**. While certain examples and examples are described below in the context of a home environment, the technologies described herein may be implemented in other types of environments. In some examples, for instance, the media playback system **100** can be implemented in one or more commercial settings (e.g., a restaurant, mall, airport, hotel, a retail or other store), one or more vehicles (e.g., a sports utility vehicle, bus, car, a ship, a boat, an airplane), multiple environments (e.g., a combination of home and vehicle environments), and/or another suitable environment where multi-zone audio may be desirable.

**[0035]** The media playback system **100** can comprise one or more playback zones, some of which may correspond to the rooms in the environment **101**. The media playback system **100** can be established with one or more playback zones, after which additional zones may be added, or removed to form, for example, the configuration shown in FIG. 1A. Each zone may be given a name according to a different room or space such as the office **101e**, master bathroom **101a**, master bedroom **101b**, the second bedroom **101c**, kitchen **101h**, dining room **101g**, living room **101f**, and/or the outdoor patio **101i**. In some aspects, a single playback zone may include multiple rooms or spaces. In certain aspects, a single room or space may include multiple playback zones.

**[0036]** In the illustrated example of FIG. 1A, the master bathroom **101a**, the second bedroom **101c**, the office **101e**, the living room **101f**, the dining room **101g**, the kitchen **101h**, and the outdoor patio **101i** each include one playback device **110**, and the master bedroom **101b** and the den **101d** include a plurality of playback devices **110**. In the master bedroom **101b**, the playback devices **110l** and **110m** may be configured, for example, to play back audio content in synchrony as individual ones of playback devices **110**, as a bonded playback zone, as a consolidated playback device, and/or any combination thereof. Similarly, in the den **101d**, the playback devices **110h-j** can be configured, for instance, to play back audio content in synchrony as individual ones of playback devices **110**, as one or more bonded playback devices, and/or as one or more consolidated playback devices. Additional details regarding bonded and consolidated playback devices are described below with respect to FIGS. 1B and 1E.

**[0037]** In some aspects, one or more of the playback zones in the environment **101** may each be playing different audio content. For instance, a user may be grilling on the patio **101i** and listening to hip hop music being played by the playback device **110c** while another user is preparing food in the kitchen **101h** and listening to classical music played by the playback device **110b**. In another example, a playback zone may play the same audio content in synchrony with another playback zone. For instance, the user may be in the office **101e** listening to the playback device **110f** playing back the same hip hop music being played back by playback

device **110c** on the patio **101i**. In some aspects, the playback devices **110c** and **110f** play back the hip hop music in synchrony such that the user perceives that the audio content is being played seamlessly (or at least substantially seamlessly) while moving between different playback zones. Additional details regarding audio playback synchronization among playback devices and/or zones can be found, for example, in U.S. Pat. No. 8,234,395 entitled, “System and method for synchronizing operations among a plurality of independently clocked digital data processing devices,” which is incorporated herein by reference in its entirety.

#### a. Suitable Media Playback System

**[0038]** FIG. 1B is a schematic diagram of the media playback system **100** and a cloud network **102**. For ease of illustration, certain devices of the media playback system **100** and the cloud network **102** are omitted from FIG. 1B. One or more communication links **103** (referred to herein after as “the links **103**”) communicatively couple the media playback system **100** and the cloud network **102**.

**[0039]** The links **103** can comprise, for example, one or more wired networks, one or more wireless networks, one or more wide area networks (WAN), one or more local area networks (LAN), one or more personal area networks (PAN), one or more telecommunication networks (e.g., one or more Global System for Mobiles (GSM) networks, Code Division Multiple Access (CDMA) networks, Long-Term Evolution (LTE) networks, 5G communication network networks, and/or other suitable data transmission protocol networks), etc. The cloud network **102** is configured to deliver media content (e.g., audio content, video content, photographs, social media content) to the media playback system **100** in response to a request transmitted from the media playback system **100** via the links **103**. In some examples, the cloud network **102** is further configured to receive data (e.g. voice input data) from the media playback system **100** and correspondingly transmit commands and/or media content to the media playback system **100**.

**[0040]** The cloud network **102** comprises computing devices **106** (identified separately as a first computing device **106a**, a second computing device **106b**, and a third computing device **106c**). The computing devices **106** can comprise individual computers or servers, such as, for example, a media streaming service server storing audio and/or other media content, a voice service server, a social media server, a media playback system control server, etc. In some examples, one or more of the computing devices **106** comprise modules of a single computer or server. In certain examples, one or more of the computing devices **106** comprise one or more modules, computers, and/or servers. Moreover, while the cloud network **102** is described above in the context of a single cloud network, in some examples the cloud network **102** comprises a plurality of cloud networks comprising communicatively coupled computing devices. Furthermore, while the cloud network **102** is shown in FIG. 1B as having three of the computing devices **106**, in some examples, the cloud network **102** comprises fewer (or more than) three computing devices **106**.

**[0041]** The media playback system **100** is configured to receive media content from the networks **102** via the links **103**. The received media content can comprise, for example, a Uniform Resource Identifier (URI) and/or a Uniform Resource Locator (URL). For instance, in some examples, the media playback system **100** can stream, download, or

otherwise obtain data from a URI or a URL corresponding to the received media content. A network **104** communicatively couples the links **103** and at least a portion of the devices (e.g., one or more of the playback devices **110**, NMDs **120**, and/or control devices **130**) of the media playback system **100**. The network **104** can include, for example, a wireless network (e.g., a WiFi network, a Bluetooth, a Z-Wave network, a ZigBee, and/or other suitable wireless communication protocol network) and/or a wired network (e.g., a network comprising Ethernet, Universal Serial Bus (USB), and/or another suitable wired communication). As those of ordinary skill in the art will appreciate, as used herein, “WiFi” can refer to several different communication protocols including, for example, Institute of Electrical and Electronics Moduleers (IEEE) 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.11ac, 802.11ad, 802.11af, 802.11ah, 802.11ai, 802.11aj, 802.11aq, 802.11ax, 802.11ay, 802.15, etc. transmitted at 2.4 Gigahertz (GHz), 5 GHz, and/or another suitable frequency.

[0042] In some examples, the network **104** comprises a dedicated communication network that the media playback system **100** uses to transmit messages between individual devices and/or to transmit media content to and from media content sources (e.g., one or more of the computing devices **106**). In certain examples, the network **104** is configured to be accessible only to devices in the media playback system **100**, thereby reducing interference and competition with other household devices. In other examples, however, the network **104** comprises an existing household communication network (e.g., a household WiFi network). In some examples, the links **103** and the network **104** comprise one or more of the same networks. In some aspects, for example, the links **103** and the network **104** comprise a telecommunication network (e.g., an LTE network, a 5G network). Moreover, in some examples, the media playback system **100** is implemented without the network **104**, and devices comprising the media playback system **100** can communicate with each other, for example, via one or more direct connections, PANs, telecommunication networks, and/or other suitable communication links.

[0043] In some examples, audio content sources may be regularly added or removed from the media playback system **100**. In some examples, for instance, the media playback system **100** performs an indexing of media items when one or more media content sources are updated, added to, and/or removed from the media playback system **100**. The media playback system **100** can scan identifiable media items in some or all folders and/or directories accessible to the playback devices **110**, and generate or update a media content database comprising metadata (e.g., title, artist, album, track length) and other associated information (e.g., URIs, URLs) for each identifiable media item found. In some examples, for instance, the media content database is stored on one or more of the playback devices **110**, NMDs **120**, and/or control devices **130**.

[0044] In the illustrated example of FIG. 1B, the playback devices **110/** and **110m** comprise a group **107a**. The playback devices **110/** and **110m** can be positioned in different rooms in a household and be grouped together in the group **107a** on a temporary or permanent basis based on user input received at the control device **130a** and/or another control device **130** in the media playback system **100**. When arranged in the group **107a**, the playback devices **110/** and **110m** can be configured to play back the same or similar audio content in

synchrony from one or more audio content sources. In certain examples, for instance, the group **107a** comprises a bonded zone in which the playback devices **110/** and **110m** comprise left audio and right audio channels, respectively, of multi-channel audio content, thereby producing or enhancing a stereo effect of the audio content. In some examples, the group **107a** includes additional playback devices **110**. In other examples, however, the media playback system **100** omits the group **107a** and/or other grouped arrangements of the playback devices **110**.

[0045] The media playback system **100** includes the NMDs **120a** and **120d**, each comprising one or more microphones configured to receive voice utterances from a user. In the illustrated example of FIG. 1B, the NMD **120a** is a standalone device and the NMD **120d** is integrated into the playback device **110n**. The NMD **120a**, for example, is configured to receive voice input **121** from a user **123**. In some examples, the NMD **120a** transmits data associated with the received voice input **121** to a voice assistant service (VAS) configured to (i) process the received voice input data and (ii) transmit a corresponding command to the media playback system **100**. In some aspects, for example, the computing device **106c** comprises one or more modules and/or servers of a VAS (e.g., a VAS operated by one or more of SONOS®, AMAZON®, GOOGLE®, APPLE®, MICROSOFT®). The computing device **106c** can receive the voice input data from the NMD **120a** via the network **104** and the links **103**. In response to receiving the voice input data, the computing device **106c** processes the voice input data (i.e., “Play Hey Jude by The Beatles”), and determines that the processed voice input includes a command to play a song (e.g., “Hey Jude”). The computing device **106c** accordingly transmits commands to the media playback system **100** to play back “Hey Jude” by the Beatles from a suitable media service (e.g., via one or more of the computing devices **106**) on one or more of the playback devices **110**.

#### b. Suitable Playback Devices

[0046] FIG. 1C is a block diagram of the playback device **110a** comprising an input/output **111**. The input/output **111** can include an analog I/O **111a** (e.g., one or more wires, cables, and/or other suitable communication links configured to carry analog signals) and/or a digital I/O **111b** (e.g., one or more wires, cables, or other suitable communication links configured to carry digital signals). In some examples, the analog I/O **111a** is an audio line-in input connection comprising, for example, an auto-detecting 3.5 mm audio line-in connection. In some examples, the digital I/O **111b** comprises a Sony/Philips Digital Interface Format (S/PDIF) communication interface and/or cable and/or a Toshiba Link (TOSLINK) cable. In some examples, the digital I/O **111b** comprises a High-Definition Multimedia Interface (HDMI) interface and/or cable. In some examples, the digital I/O **111b** includes one or more wireless communication links comprising, for example, a radio frequency (RF), infrared, WiFi, Bluetooth, or another suitable communication protocol. In certain examples, the analog I/O **111a** and the digital **111b** comprise interfaces (e.g., ports, plugs, jacks) configured to receive connectors of cables transmitting analog and digital signals, respectively, without necessarily including cables.

[0047] The playback device **110a**, for example, can receive media content (e.g., audio content comprising music

and/or other sounds) from a local audio source **105** via the input/output **111** (e.g., a cable, a wire, a PAN, a Bluetooth connection, an ad hoc wired or wireless communication network, and/or another suitable communication link). The local audio source **105** can comprise, for example, a mobile device (e.g., a smartphone, a tablet, a laptop computer) or another suitable audio component (e.g., a television, a desktop computer, an amplifier, a phonograph, a Blu-ray player, a memory storing digital media files). In some aspects, the local audio source **105** includes local music libraries on a smartphone, a computer, a networked-attached storage (NAS), and/or another suitable device configured to store media files. In certain examples, one or more of the playback devices **110**, NMDs **120**, and/or control devices **130** comprise the local audio source **105**. In other examples, however, the media playback system omits the local audio source **105** altogether. In some examples, the playback device **110a** does not include an input/output **111** and receives all audio content via the network **104**.

**[0048]** The playback device **110a** further comprises electronics **112**, a user interface **113** (e.g., one or more buttons, knobs, dials, touch-sensitive surfaces, displays, touchscreens), and one or more transducers **114** (referred to hereinafter as “the transducers **114**”). The electronics **112** is configured to receive audio from an audio source (e.g., the local audio source **105**) via the input/output **111**, one or more of the computing devices **106a-c** via the network **104** (FIG. 1B)), amplify the received audio, and output the amplified audio for playback via one or more of the transducers **114**. In some examples, the playback device **110a** optionally includes one or more microphones **115** (e.g., a single microphone, a plurality of microphones, a microphone array) (hereinafter referred to as “the microphones **115**”). In certain examples, for instance, the playback device **110a** having one or more of the optional microphones **115** can operate as an NMD configured to receive voice input from a user and correspondingly perform one or more operations based on the received voice input.

**[0049]** In the illustrated example of FIG. 1C, the electronics **112** comprise one or more processors **112a** (referred to hereinafter as “the processors **112a**”), memory **112b**, software components **112c**, a network interface **112d**, one or more audio processing components **112g** (referred to hereinafter as “the audio components **112g**”), one or more audio amplifiers **112h** (referred to hereinafter as “the amplifiers **112h**”), and power **112i** (e.g., one or more power supplies, power cables, power receptacles, batteries, induction coils, Power-over Ethernet (POE) interfaces, and/or other suitable sources of electric power). In some examples, the electronics **112** optionally include one or more other components **112j** (e.g., one or more sensors, video displays, touchscreens, battery charging bases).

**[0050]** The processors **112a** can comprise clock-driven computing component(s) configured to process data, and the memory **112b** can comprise a computer-readable medium (e.g., a tangible, non-transitory computer-readable medium, data storage loaded with one or more of the software components **112c**) configured to store instructions for performing various operations and/or functions. The processors **112a** are configured to execute the instructions stored on the memory **112b** to perform one or more of the operations. The operations can include, for example, causing the playback device **110a** to retrieve audio data from an audio source (e.g., one or more of the computing devices **106a-c** (FIG.

1B)), and/or another one of the playback devices **110**. In some examples, the operations further include causing the playback device **110a** to send audio data to another one of the playback devices **110a** and/or another device (e.g., one of the NMDs **120**). Certain examples include operations causing the playback device **110a** to pair with another of the one or more playback devices **110** to enable a multi-channel audio environment (e.g., a stereo pair, a bonded zone).

**[0051]** The processors **112a** can be further configured to perform operations causing the playback device **110a** to synchronize playback of audio content with another of the one or more playback devices **110**. As those of ordinary skill in the art will appreciate, during synchronous playback of audio content on a plurality of playback devices, a listener will preferably be unable to perceive time-delay differences between playback of the audio content by the playback device **110a** and the other one or more other playback devices **110**. Additional details regarding audio playback synchronization among playback devices can be found, for example, in U.S. Pat. No. 8,234,395, which was incorporated by reference above.

**[0052]** In some examples, the memory **112b** is further configured to store data associated with the playback device **110a**, such as one or more zones and/or zone groups of which the playback device **110a** is a member, audio sources accessible to the playback device **110a**, and/or a playback queue that the playback device **110a** (and/or another of the one or more playback devices) can be associated with. The stored data can comprise one or more state variables that are periodically updated and used to describe a state of the playback device **110a**. The memory **112b** can also include data associated with a state of one or more of the other devices (e.g., the playback devices **110**, NMDs **120**, control devices **130**) of the media playback system **100**. In some aspects, for example, the state data is shared during predetermined intervals of time (e.g., every 5 seconds, every 10 seconds, every 60 seconds) among at least a portion of the devices of the media playback system **100**, so that one or more of the devices have the most recent data associated with the media playback system **100**.

**[0053]** The network interface **112d** is configured to facilitate a transmission of data between the playback device **110a** and one or more other devices on a data network such as, for example, the links **103** and/or the network **104** (FIG. 1B). The network interface **112d** is configured to transmit and receive data corresponding to media content (e.g., audio content, video content, text, photographs) and other signals (e.g., non-transitory signals) comprising digital packet data including an Internet Protocol (IP)-based source address and/or an IP-based destination address. The network interface **112d** can parse the digital packet data such that the electronics **112** properly receives and processes the data destined for the playback device **110a**.

**[0054]** In the illustrated example of FIG. 1C, the network interface **112d** comprises one or more wireless interfaces **112e** (referred to hereinafter as “the wireless interface **112e**”). The wireless interface **112e** (e.g., a suitable interface comprising one or more antennae) can be configured to wirelessly communicate with one or more other devices (e.g., one or more of the other playback devices **110**, NMDs **120**, and/or control devices **130**) that are communicatively coupled to the network **104** (FIG. 1B) in accordance with a suitable wireless communication protocol (e.g., WiFi, Bluetooth, LTE). In some examples, the network interface **112d**

optionally includes a wired interface **112f** (e.g., an interface or receptacle configured to receive a network cable such as an Ethernet, a USB-A, USB-C, and/or Thunderbolt cable) configured to communicate over a wired connection with other devices in accordance with a suitable wired communication protocol. In certain examples, the network interface **112d** includes the wired interface **112f** and excludes the wireless interface **112e**. In some examples, the electronics **112** excludes the network interface **112d** altogether and transmits and receives media content and/or other data via another communication path (e.g., the input/output **111**).

**[0055]** The audio components **112g** are configured to process and/or filter data comprising media content received by the electronics **112** (e.g., via the input/output **111** and/or the network interface **112d**) to produce output audio signals. In some examples, the audio processing components **112g** comprise, for example, one or more digital-to-analog converters (DAC), audio preprocessing components, audio enhancement components, a digital signal processors (DSPs), and/or other suitable audio processing components, modules, circuits, etc. In certain examples, one or more of the audio processing components **112g** can comprise one or more subcomponents of the processors **112a**. In some examples, the electronics **112** omits the audio processing components **112g**. In some aspects, for example, the processors **112a** execute instructions stored on the memory **112b** to perform audio processing operations to produce the output audio signals.

**[0056]** The amplifiers **112h** are configured to receive and amplify the audio output signals produced by the audio processing components **112g** and/or the processors **112a**. The amplifiers **112h** can comprise electronic devices and/or components configured to amplify audio signals to levels sufficient for driving one or more of the transducers **114**. In some examples, for instance, the amplifiers **112h** include one or more switching or class-D power amplifiers. In other examples, however, the amplifiers include one or more other types of power amplifiers (e.g., linear gain power amplifiers, class-A amplifiers, class-B amplifiers, class-AB amplifiers, class-C amplifiers, class-D amplifiers, class-E amplifiers, class-F amplifiers, class-G and/or class H amplifiers, and/or another suitable type of power amplifier). In certain examples, the amplifiers **112h** comprise a suitable combination of two or more of the foregoing types of power amplifiers. Moreover, in some examples, individual ones of the amplifiers **112h** correspond to individual ones of the transducers **114**. In other examples, however, the electronics **112** includes a single one of the amplifiers **112h** configured to output amplified audio signals to a plurality of the transducers **114**. In some other examples, the electronics **112** omits the amplifiers **112h**.

**[0057]** The transducers **114** (e.g., one or more speakers and/or speaker drivers) receive the amplified audio signals from the amplifier **112h** and render or output the amplified audio signals as sound (e.g., audible sound waves having a frequency between about 20 Hertz (Hz) and 20 kilohertz (kHz)). In some examples, the transducers **114** can comprise a single transducer. In other examples, however, the transducers **114** comprise a plurality of audio transducers. In some examples, the transducers **114** comprise more than one type of transducer. For example, the transducers **114** can include one or more low frequency transducers (e.g., subwoofers, woofers), mid-range frequency transducers (e.g., mid-range transducers, mid-woofers), and one or more high

frequency transducers (e.g., one or more tweeters). As used herein, “low frequency” can generally refer to audible frequencies below about 500 Hz, “mid-range frequency” can generally refer to audible frequencies between about 500 Hz and about 2 kHz, and “high frequency” can generally refer to audible frequencies above 2 kHz. In certain examples, however, one or more of the transducers **114** comprise transducers that do not adhere to the foregoing frequency ranges. For example, one of the transducers **114** may comprise a mid-woofer transducer configured to output sound at frequencies between about 200 Hz and about 5 kHz.

**[0058]** By way of illustration, SONOS, Inc. presently offers (or has offered) for sale certain playback devices including, for example, a “SONOS ONE,” “PLAY:1,” “PLAY:3,” “PLAY:5,” “PLAYBAR,” “PLAYBASE,” “CONNECT:AMP,” “CONNECT,” and “SUB.” Other suitable playback devices may additionally or alternatively be used to implement the playback devices of example examples disclosed herein. Additionally, one of ordinary skilled in the art will appreciate that a playback device is not limited to the examples described herein or to SONOS product offerings. In some examples, for instance, one or more playback devices **110** comprises wired or wireless headphones (e.g., over-the-ear headphones, on-ear headphones, in-ear earphones). In other examples, one or more of the playback devices **110** comprise a docking station and/or an interface configured to interact with a docking station for personal mobile media playback devices. In certain examples, a playback device may be integral to another device or component such as a television, a lighting fixture, or some other device for indoor or outdoor use. In some examples, a playback device omits a user interface and/or one or more transducers. For example, FIG. 1D is a block diagram of a playback device **110p** comprising the input/output **111** and electronics **112** without the user interface **113** or transducers **114**.

**[0059]** FIG. 1E is a block diagram of a bonded playback device **110q** comprising the playback device **110a** (FIG. 1C) sonically bonded with the playback device **110i** (e.g., a subwoofer) (FIG. 1A). In the illustrated example, the playback devices **110a** and **110i** are separate ones of the playback devices **110** housed in separate enclosures. In some examples, however, the bonded playback device **110q** comprises a single enclosure housing both the playback devices **110a** and **110i**. The bonded playback device **110q** can be configured to process and reproduce sound differently than an unbonded playback device (e.g., the playback device **110a** of FIG. 1C) and/or paired or bonded playback devices (e.g., the playback devices **110l** and **110m** of FIG. 1B). In some examples, for instance, the playback device **110a** is full-range playback device configured to render low frequency, mid-range frequency, and high frequency audio content, and the playback device **110i** is a subwoofer configured to render low frequency audio content. In some aspects, the playback device **110a**, when bonded with the first playback device, is configured to render only the mid-range and high frequency components of a particular audio content, while the playback device **110i** renders the low frequency component of the particular audio content. In some examples, the bonded playback device **110q** includes additional playback devices and/or another bonded playback device.

### c. Suitable Network Microphone Devices (NMDs)

**[0060]** FIG. 1F is a block diagram of the NMD **120a** (FIGS. 1A and 1B). The NMD **120a** includes one or more voice processing components **124** (hereinafter “the voice components **124**”) and several components described with respect to the playback device **110a** (FIG. 1C) including the processors **112a**, the memory **112b**, and the microphones **115**. The NMD **120a** optionally comprises other components also included in the playback device **110a** (FIG. 1C), such as the user interface **113** and/or the transducers **114**. In some examples, the NMD **120a** is configured as a media playback device (e.g., one or more of the playback devices **110**), and further includes, for example, one or more of the audio components **112g** (FIG. 1C), the amplifiers **114**, and/or other playback device components. In certain examples, the NMD **120a** comprises an Internet of Things (IoT) device such as, for example, a thermostat, alarm panel, fire and/or smoke detector, etc. In some examples, the NMD **120a** comprises the microphones **115**, the voice processing **124**, and only a portion of the components of the electronics **112** described above with respect to FIG. 1B. In some aspects, for example, the NMD **120a** includes the processor **112a** and the memory **112b** (FIG. 1B), while omitting one or more other components of the electronics **112**. In some examples, the NMD **120a** includes additional components (e.g., one or more sensors, cameras, thermometers, barometers, hygrometers). **[0061]** In some examples, an NMD can be integrated into a playback device. FIG. 1G is a block diagram of a playback device **110r** comprising an NMD **120d**. The playback device **110r** can comprise many or all of the components of the playback device **110a** and further include the microphones **115** and voice processing **124** (FIG. 1F). The playback device **110r** optionally includes an integrated control device **130c**. The control device **130c** can comprise, for example, a user interface (e.g., the user interface **113** of FIG. 1B) configured to receive user input (e.g., touch input, voice input) without a separate control device. In other examples, however, the playback device **110r** receives commands from another control device (e.g., the control device **130a** of FIG. 1B).

**[0062]** Referring again to FIG. 1F, the microphones **115** are configured to acquire, capture, and/or receive sound from an environment (e.g., the environment **101** of FIG. 1A) and/or a room in which the NMD **120a** is positioned. The received sound can include, for example, vocal utterances, audio played back by the NMD **120a** and/or another playback device, background voices, ambient sounds, etc. The microphones **115** convert the received sound into electrical signals to produce microphone data. The voice processing **124** receives and analyzes the microphone data to determine whether a voice input is present in the microphone data. The voice input can comprise, for example, an activation word followed by an utterance including a user request. As those of ordinary skill in the art will appreciate, an activation word is a word or other audio cue that signifying a user voice input. For instance, in querying the AMAZON® VAS, a user might speak the activation word “Alexa.” Other examples include “Ok, Google” for invoking the GOOGLE® VAS and “Hey, Siri” for invoking the APPLE® VAS.

**[0063]** After detecting the activation word, voice processing **124** monitors the microphone data for an accompanying user request in the voice input. The user request may include, for example, a command to control a third-party device, such as a thermostat (e.g., NEST® thermostat), an

illumination device (e.g., a PHILIPS HUE® lighting device), or a media playback device (e.g., a Sonos® playback device). For example, a user might speak the activation word “Alexa” followed by the utterance “set the thermostat to 68 degrees” to set a temperature in a home (e.g., the environment **101** of FIG. 1A). The user might speak the same activation word followed by the utterance “turn on the living room” to turn on illumination devices in a living room area of the home. The user may similarly speak an activation word followed by a request to play a particular song, an album, or a playlist of music on a playback device in the home.

### d. Suitable Control Devices

**[0064]** FIG. 1H is a partially schematic diagram of the control device **130a** (FIGS. 1A and 1B). As used herein, the term “control device” can be used interchangeably with “controller” or “control system.” Among other features, the control device **130a** is configured to receive user input related to the media playback system **100** and, in response, cause one or more devices in the media playback system **100** to perform an action(s) or operation(s) corresponding to the user input. In the illustrated example, the control device **130a** comprises a smartphone (e.g., an iPhone™, an Android phone) on which media playback system controller application software is installed. In some examples, the control device **130a** comprises, for example, a tablet (e.g., an iPad™), a computer (e.g., a laptop computer, a desktop computer), and/or another suitable device (e.g., a television, an automobile audio head unit, an IoT device). In certain examples, the control device **130a** comprises a dedicated controller for the media playback system **100**. In other examples, as described above with respect to FIG. 1G, the control device **130a** is integrated into another device in the media playback system **100** (e.g., one more of the playback devices **110**, NMDs **120**, and/or other suitable devices configured to communicate over a network).

**[0065]** The control device **130a** includes electronics **132**, a user interface **133**, one or more speakers **134**, and one or more microphones **135**. The electronics **132** comprise one or more processors **132a** (referred to hereinafter as “the processors **132a**”), a memory **132b**, software components **132c**, and a network interface **132d**. The processor **132a** can be configured to perform functions relevant to facilitating user access, control, and configuration of the media playback system **100**. The memory **132b** can comprise data storage that can be loaded with one or more of the software components executable by the processor **112a** to perform those functions. The software components **132c** can comprise applications and/or other executable software configured to facilitate control of the media playback system **100**. The memory **112b** can be configured to store, for example, the software components **132c**, media playback system controller application software, and/or other data associated with the media playback system **100** and the user.

**[0066]** The network interface **132d** is configured to facilitate network communications between the control device **130a** and one or more other devices in the media playback system **100**, and/or one or more remote devices. In some examples, the network interface **132d** is configured to operate according to one or more suitable communication industry standards (e.g., infrared, radio, wired standards including IEEE 802.3, wireless standards including IEEE 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.15, 4G, LTE). The

network interface **132d** can be configured, for example, to transmit data to and/or receive data from the playback devices **110**, the NMDs **120**, other ones of the control devices **130**, one of the computing devices **106** of FIG. 1B, devices comprising one or more other media playback systems, etc. The transmitted and/or received data can include, for example, playback device control commands, state variables, playback zone and/or zone group configurations. For instance, based on user input received at the user interface **133**, the network interface **132d** can transmit a playback device control command (e.g., volume control, audio playback control, audio content selection) from the control device **130** to one or more of the playback devices **110**. The network interface **132d** can also transmit and/or receive configuration changes such as, for example, adding/removing one or more playback devices **110** to/from a zone, adding/removing one or more zones to/from a zone group, forming a bonded or consolidated player, separating one or more playback devices from a bonded or consolidated player, among others. Additional description of zones and groups can be found below with respect to FIGS. 1I through 1M.

**[0067]** The user interface **133** is configured to receive user input and can facilitate control of the media playback system **100**. The user interface **133** includes media content art **133a** (e.g., album art, lyrics, videos), a playback status indicator **133b** (e.g., an elapsed and/or remaining time indicator), media content information region **133c**, a playback control region **133d**, and a zone indicator **133e**. The media content information region **133c** can include a display of relevant information (e.g., title, artist, album, genre, release year) about media content currently playing and/or media content in a queue or playlist. The playback control region **133d** can include selectable (e.g., via touch input and/or via a cursor or another suitable selector) icons to cause one or more playback devices in a selected playback zone or zone group to perform playback actions such as, for example, play or pause, fast forward, rewind, skip to next, skip to previous, enter/exit shuffle mode, enter/exit repeat mode, enter/exit cross fade mode, etc. The playback control region **133d** may also include selectable icons to modify equalization settings, playback volume, and/or other suitable playback actions. In the illustrated example, the user interface **133** comprises a display presented on a touch screen interface of a smartphone (e.g., an iPhone™, an Android phone). In some examples, however, user interfaces of varying formats, styles, and interactive sequences may alternatively be implemented on one or more network devices to provide comparable control access to a media playback system.

**[0068]** The one or more speakers **134** (e.g., one or more transducers) can be configured to output sound to the user of the control device **130a**. In some examples, the one or more speakers comprise individual transducers configured to correspondingly output low frequencies, mid-range frequencies, and/or high frequencies. In some aspects, for example, the control device **130a** is configured as a playback device (e.g., one of the playback devices **110**). Similarly, in some examples the control device **130a** is configured as an NMD (e.g., one of the NMDs **120**), receiving voice commands and other sounds via the one or more microphones **135**.

**[0069]** The one or more microphones **135** can comprise, for example, one or more condenser microphones, electret condenser microphones, dynamic microphones, and/or other

suitable types of microphones or transducers. In some examples, two or more of the microphones **135** are arranged to capture location information of an audio source (e.g., voice, audible sound) and/or configured to facilitate filtering of background noise. Moreover, in certain examples, the control device **130a** is configured to operate as playback device and an NMD. In other examples, however, the control device **130a** omits the one or more speakers **134** and/or the one or more microphones **135**. For instance, the control device **130a** may comprise a device (e.g., a thermostat, an IoT device, a network device) comprising a portion of the electronics **132** and the user interface **133** (e.g., a touch screen) without any speakers or microphones.

#### Suitable Playback Device Configurations

**[0070]** FIGS. 1I through 1M show example configurations of playback devices in zones and zone groups. Referring first to FIG. 1M, in one example, a single playback device may belong to a zone. For example, the playback device **110g** in the second bedroom **101c** (FIG. 1A) may belong to Zone C. In some implementations described below, multiple playback devices may be “bonded” to form a “bonded pair” which together form a single zone. For example, the playback device **110l** (e.g., a left playback device) can be bonded to the playback device **110j** (e.g., a left playback device) to form Zone A. Bonded playback devices may have different playback responsibilities (e.g., channel responsibilities). In another implementation described below, multiple playback devices may be merged to form a single zone. For example, the playback device **110h** (e.g., a front playback device) may be merged with the playback device **110i** (e.g., a subwoofer), and the playback devices **110j** and **110k** (e.g., left and right surround speakers, respectively) to form a single Zone D. In another example, the playback devices **110g** and **110h** can be merged to form a merged group or a zone group **108b**. The merged playback devices **110g** and **110h** may not be specifically assigned different playback responsibilities. That is, the merged playback devices **110h** and **110i** may, aside from playing audio content in synchrony, each play audio content as they would if they were not merged.

**[0071]** Each zone in the media playback system **100** may be provided for control as a single user interface (UI) entity. For example, Zone A may be provided as a single entity named Master Bathroom. Zone B may be provided as a single entity named Master Bedroom. Zone C may be provided as a single entity named Second Bedroom.

**[0072]** Playback devices that are bonded may have different playback responsibilities, such as responsibilities for certain audio channels. For example, as shown in FIG. 1-I, the playback devices **110l** and **110m** may be bonded so as to produce or enhance a stereo effect of audio content. In this example, the playback device **110l** may be configured to play a left channel audio component, while the playback device **110k** may be configured to play a right channel audio component. In some implementations, such stereo bonding may be referred to as “pairing.”

**[0073]** Additionally, bonded playback devices may have additional and/or different respective speaker drivers. As shown in FIG. 1J, the playback device **110h** named Front may be bonded with the playback device **110i** named SUB. The Front device **110h** can be configured to render a range of mid to high frequencies and the SUB device **110i** can be configured render low frequencies. When unbonded, however, the Front device **110h** can be configured render a full

range of frequencies. As another example, FIG. 1K shows the Front and SUB devices **110h** and **110i** further bonded with Left and Right playback devices **110j** and **110k**, respectively. In some implementations, the Right and Left devices **110j** and **110k** can be configured to form surround or “satellite” channels of a home theater system. The bonded playback devices **110h**, **110i**, **110j**, and **110k** may form a single Zone D (FIG. 1M).

**[0074]** Playback devices that are merged may not have assigned playback responsibilities, and may each render the full range of audio content the respective playback device is capable of. Nevertheless, merged devices may be represented as a single UI entity (i.e., a zone, as discussed above). For instance, the playback devices **110a** and **110n** the master bathroom have the single UI entity of Zone A. In one example, the playback devices **110a** and **110n** may each output the full range of audio content each respective playback devices **110a** and **110n** are capable of, in synchrony.

**[0075]** In some examples, an NMD is bonded or merged with another device so as to form a zone. For example, the NMD **120b** may be bonded with the playback device **110e**, which together form Zone F, named Living Room. In other examples, a stand-alone network microphone device may be in a zone by itself. In other examples, however, a stand-alone network microphone device may not be associated with a zone. Additional details regarding associating network microphone devices and playback devices as designated or default devices may be found, for example, in previously referenced U.S. patent application Ser. No. 15/438,749.

**[0076]** Zones of individual, bonded, and/or merged devices may be grouped to form a zone group. For example, referring to FIG. 1M, Zone A may be grouped with Zone B to form a zone group **108a** that includes the two zones. Similarly, Zone G may be grouped with Zone H to form the zone group **108b**. As another example, Zone A may be grouped with one or more other Zones C-I. The Zones A-I may be grouped and ungrouped in numerous ways. For example, three, four, five, or more (e.g., all) of the Zones A-I may be grouped. When grouped, the zones of individual and/or bonded playback devices may play back audio in synchrony with one another, as described in previously referenced U.S. Pat. No. 8,234,395. Playback devices may be dynamically grouped and ungrouped to form new or different groups that synchronously play back audio content.

**[0077]** In various implementations, the zones in an environment may be the default name of a zone within the group or a combination of the names of the zones within a zone group. For example, Zone Group **108b** can have be assigned a name such as “Dining +Kitchen”, as shown in FIG. 1M. In some examples, a zone group may be given a unique name selected by a user.

**[0078]** Certain data may be stored in a memory of a playback device (e.g., the memory **112b** of FIG. 1C) as one or more state variables that are periodically updated and used to describe the state of a playback zone, the playback device(s), and/or a zone group associated therewith. The memory may also include the data associated with the state of the other devices of the media system, and shared from time to time among the devices so that one or more of the devices have the most recent data associated with the system.

**[0079]** In some examples, the memory may store instances of various variable types associated with the states. Variables

instances may be stored with identifiers (e.g., tags) corresponding to type. For example, certain identifiers may be a first type “a1” to identify playback device(s) of a zone, a second type “b1” to identify playback device(s) that may be bonded in the zone, and a third type “c1” to identify a zone group to which the zone may belong. As a related example, identifiers associated with the second bedroom **101c** may indicate that the playback device is the only playback device of the Zone C and not in a zone group. Identifiers associated with the Den may indicate that the Den is not grouped with other zones but includes bonded playback devices **110h-110k**. Identifiers associated with the Dining Room may indicate that the Dining Room is part of the Dining+Kitchen zone group **108b** and that devices **110b** and **110d** are grouped (FIG. 1L). Identifiers associated with the Kitchen may indicate the same or similar information by virtue of the Kitchen being part of the Dining+Kitchen zone group **108b**. Other example zone variables and identifiers are described below.

**[0080]** In yet another example, the media playback system **100** may variables or identifiers representing other associations of zones and zone groups, such as identifiers associated with Areas, as shown in FIG. 1M. An area may involve a cluster of zone groups and/or zones not within a zone group. For instance, FIG. 1M shows an Upper Area **109a** including Zones A-D, and a Lower Area **109b** including Zones E-I. In one aspect, an Area may be used to invoke a cluster of zone groups and/or zones that share one or more zones and/or zone groups of another cluster. In another aspect, this differs from a zone group, which does not share a zone with another zone group. Further examples of techniques for implementing Areas may be found, for example, in U.S. application Ser. No. 15/682,506 filed Aug. 21, 2017 and titled “Room Association Based on Name,” and U.S. Pat. No. 8,483,853 filed Sep. 11, 2007, and titled “Controlling and manipulating groupings in a multi-zone media system.” Each of these applications is incorporated herein by reference in its entirety. In some examples, the media playback system **100** may not implement Areas, in which case the system may not store variables associated with Areas.

### III. Multi-Device Playback of Generative Media Content

**[0081]** FIG. 2 is a functional block diagram of a system **200** for playback of generative media content. As noted previously, generative media content can include any media content (e.g., audio, video, audio-visual output, tactile output, or any other media content) that is dynamically created, synthesized, and/or modified by a non-human, rule-based process such as an algorithm or model. This creation or modification can occur for playback in real-time or near real-time. Additionally or alternatively, generative media content can be produced or modified asynchronously (e.g., ahead of time before playback is requested), and the particular item of generative media content may then be selected for playback at a later time. As used herein, a “generative media module” includes any system, whether implemented in software, a physical model, or combination thereof, that can produce generative media content based on one or more inputs. In some examples, such generative media content includes novel media content that can be created as wholly new or can be created by mixing, combining, manipulating, or otherwise modifying one or more pre-existing pieces of media content. As used herein, a

“generative media content model” includes any algorithm, schema, or set of rules that can be used to produce novel generative media content using one or more inputs (e.g., sensor data, artist-provided parameters, media segments such as audio clips or samples, etc.). Among examples, a generative media module can use a variety of different generative media content models to produce different generative media content. In some instances, artists or other collaborators can interact with, author, and/or update generative media content models to produce particular generative media content. Although several examples throughout this discussion refer to audio content, the principles disclosed herein can be applied in some examples to other types of media content, e.g., video, audio-visual, tactile, or otherwise.

**[0082]** As shown in FIG. 2, the system 200 includes a generative media group coordinator 210, which is in communication with generative media group members 250a and 250b, as well as with sensor data source(s) 218, media content source(s) 220, and a control device 130. Such communication can be carried out via network(s) 102, which as noted above can include any suitable wired or wireless network connections or combinations thereof (e.g., WiFi network, a Bluetooth, a Z-Wave network, a ZigBee, an Ethernet connection, a Universal Serial Bus (USB) connection, etc.).

**[0083]** One or more remote computing device(s) 106 can also be in communication with the group coordinator 210 and/or group members 250a and 250b via the network(s) 102. In various examples, the remote computing device(s) 106 can be cloud-based servers associated with a device manufacturer, media content provider, voice-assistant service, or other suitable entity. As shown in FIG. 2, the remote computing device(s) 106 can include a generative media module 214. As described in more detail elsewhere herein, the remote computing device(s) 106 can produce generative media content remotely from the local devices (e.g., coordinator 210 and members 250a and 250b). The generative media content can then be transmitted to one or more of the local devices for playback. Additionally or alternatively, the generative media content can be produced wholly or in part via the local devices (e.g., group coordinator 210 and/or group members 250a and 250b). In some examples, the group coordinator 210 can itself be a remote computing device such that it is communicatively coupled to the group members 250a and 250b via a wide area network and the devices need not be co-located within the same environment (e.g., household, business location, etc.).

#### a. Example Generative Media Group Operation

**[0084]** In the illustrated example, a generative media group includes a generative media group coordinator 210 (also referred to herein as a “coordinator device 210”) and first and second generative media group members 250a and 250b (also referred to herein as “first member device 250a,” “second member device 250b,” and, collectively, “member devices 250”). Optionally, one or more remote computing devices 106 can also form part of the generative media group. In operation, these devices can communicate with one another and/or with other components (e.g., sensor data source(s) 218, control device 130, media content source(s) 220, or any other suitable data sources or components) to facilitate the production and playback of generative media content.

**[0085]** In various examples, some or all of the devices 210 and/or 250 can be co-located within the same environment (e.g., within the same household, store, etc.). In some examples, at least some of the devices 210 and/or 250 can be remote from one another, for example within different households, different cities, etc.

**[0086]** The coordinator device 210 and/or the member devices 250 can include some or all of the components of the playback device 110 or network microphone device 120 described above with respect to FIGS. 1A-1H. For example, the coordinator device 210 and/or member devices 250 can optionally include playback components 212 (e.g., transducers, amplifiers, audio processing components, etc.), or such components can be omitted in some instances.

**[0087]** In some examples, the coordinator device 210 is a playback device itself, and as such may also operate as a member device 250. In other examples, the coordinator device 210 can be connected to one or more member devices 250 (e.g., via direct wired connection or via network 102) but the coordinator device 210 does not itself play back generative media content. In various examples, the coordinator device 210 can be implemented on a bridge-like device on a local network, on a playback device that is not itself part of the generative media group (i.e., the playback device does not itself play back the generative media content), and/or on a remote computing device (e.g., a cloud server).

**[0088]** In various examples, one or more of the devices can include a generative media module 214 thereon. Such generative media module(s) 214 can produce novel, synthetic media content based on one or more inputs, for example using a suitable generative media content model. As shown in FIG. 2, in some examples the coordinator device 210 can include a generative media module 214 for producing generative media content, which can then be transmitted to the member devices 250a and 250b for concurrent and/or synchronous playback. Additionally or alternatively, some or all of the member devices 250 (e.g., member device 250b as shown in FIG. 2) can include a generative media module 214, which can be used by the member device 250 to locally produce generative media content based on one or more inputs. In various examples, the generative media content can be produced via the remote computing device(s) 106, optionally using one or more input parameters received from local devices. This generative media content can then be transmitted to one or more of the local devices for coordination and/or playback.

**[0089]** In some examples, at least some of the member devices 250 do not include a generative media module 214 thereon. Alternatively, in some instances each member device 250 can include a generative media module 214 thereon, and can be configured to produce generative media content locally. In at least some examples, none of the member devices 250 include a generative media module 214 thereon. In such cases, generative media content can be produced by the coordinator device 210. Such generative media content can then be transmitted to the member devices 250 for concurrent and/or synchronous playback.

**[0090]** In the example shown in FIG. 2, the coordinator device 210 additionally includes coordination components 216. As described in more detail herein, in some instances the coordinator device 210 can facilitate playback of generative media content via multiple different playback devices (which may or may not include the coordinator device 210 itself). In operation, the coordination compo-

nents **216** are configured to facilitate synchronization of both generative media creation (e.g., using one or more generative media modules **214**, which may be distributed among the various devices) as well as generative media playback. For example, the coordinator device **210** can transmit timing data to the member devices **250** to facilitate synchronous playback. Additionally or alternatively, the coordinator device **210** can transmit inputs, generative media model parameters, or other data relating to the generative media module **214** to one or more member devices **250** such that the member devices **250** can produce generative media locally (e.g., using a locally stored generative media module **214**), and/or such that the member devices **250** can update or modify the generative media modules **214** based on the inputs received from the coordinator device **210**.

[0091] As described in more detail elsewhere herein, the generative media module(s) **214** can be configured to produce generative media based on one or more inputs using a generative media content model. The inputs can include sensor data (e.g., as provided by sensor data source(s) **218**), user input (e.g., as received from control device **130** or via direct user interaction with the coordinator device **210** or member devices **250**), and/or media content source(s) **220**. For example, a generative media module **214** can produce and continuously modify generative audio by adjusting various characteristics of the generative audio based on one or more input parameters (e.g., sensor data relating to one or more users relative to the devices **210**, **250**).

#### b. Example Media Content Source(s)

[0092] The media content source(s) **220** can include, in various examples, one or more local and/or remote media content sources. For example, the media content source(s) **220** can include one or more local audio sources **105** as described above (e.g., audio received over an input/output connection such as from a mobile device (e.g., a smartphone, a tablet, a laptop computer) or another suitable audio component (e.g., a television, a desktop computer, an amplifier, a phonograph, a Blu-ray player, a memory storing digital media files)). Additionally or alternatively, the media content source(s) **220** can include one or more remote computing devices accessible via a network interface (e.g., via communication over the network(s) **102**). Such remote computing devices can include individual computers or servers, such as, for example, a media streaming service server storing audio and/or other media content, etc.

[0093] In various examples, the media available via the media content source(s) **220** can include pre-recorded audio segments in the form of complete sounds, songs, portions of songs (e.g., samples), or any audio component (e.g., pre-recorded audio of a particular instrument, synthetic beats or other audio segments, non-musical audio such as spoken word or nature sounds, etc.). In operation, such media can be utilized by the generative media modules **214** to produce generative media content, for example by combining, mixing, overlapping, manipulating, or otherwise modifying the retrieved media content to produce novel generative media content for playback via one or more devices. In some examples, the generative media content can take the form of a combination of pre-recorded audio segments (e.g., a pre-recorded song, spoken word recording, etc.) with novel, synthesized audio being created and overlaid with the pre-

recorded audio. As used herein, “generative media content” or “generated media content” can include any such combination.

#### c. Example Generative Media Modules

[0094] As noted above, the generative media module **214** can include any system, whether instantiated in software, a physical model, or combination thereof, that can produce generative media content based on one or more inputs. In various examples, the generative media module **214** can utilize a generative media content model, which can include one or more algorithms or mathematical models that determine the manner in which media content is generated based on the relevant input parameters. In some instances, the algorithms and/or mathematical models can themselves be updated over time, for example based on instructions received from one or more remote computing devices (e.g., cloud servers associated with a music service or other entity), or based on inputs received from other group member devices within the same or a different environment, or any other suitable input. In some examples, various devices within the group can have different generative media modules **214** thereon—for example with a first member device having a different generative media module **214** than a second member device. In other cases, each device within the group that has a generative media module **214** can include substantially the same model or algorithm.

[0095] Any suitable algorithm or combination of algorithms can be used to produce generative media content. Examples of such algorithms include those using machine learning techniques (e.g., generative adversarial networks, neural networks, etc.), formal grammars, Markov models, finite-state automata, and/or any algorithms implemented within currently available offerings such as JukeBox by OpenAI, AWS DeepComposer by Amazon, Magenta by Google, AmperAI by Amper Music, etc. In various examples, the generative media module(s) **214** can utilize any suitable generative algorithms now existing or developed in the future.

[0096] In line with the discussion above, producing the generative media content (e.g., audio content) can involve changing various characteristics of the media content in real time and/or algorithmically generating novel media content in real-time or near real-time. In the context of audio content, this can be achieved by storing a number of audio samples in a database (e.g., within media content source(s) **220**) that can be remotely located and accessible by the coordinator device **210** and/or the member devices **250** over the network(s) **102**, or alternatively the audio samples can be locally maintained on the devices **210**, **250** themselves. The audio samples can be associated with one or more metadata tags corresponding to one or more audio characteristics of the samples. For instance, a given sample can be associated with metadata tags indicating that the sample contains audio of a particular frequency or frequency range (e.g., bass/midrange/treble) or a particular instrument, genre, tempo, key, release date, geographical region, timbre, reverb, distortion, sonic texture, or any other audio characteristics that will be apparent.

[0097] In operation, the generative media modules **214** (e.g., of the coordinator device **210** and/or the second member device **250b**) can retrieve certain audio samples based on their associated tags and mix the audio samples together to create the generative audio. The generative audio

can evolve in real time as the generative media module(s) **214** retrieve audio samples with different tags and/or different audio samples with the same or similar tags. The audio samples that the generative media module(s) **214** retrieve can depend on one or more inputs, such as sensor data, time of day, geographic location, weather, or various user inputs, such as mood selection or physiological inputs such as heart rate or the like. In this manner, as the inputs change, so too does the generative audio. For example, if a user selects a calming or relaxation mood input, then the generative media module(s) **214** can retrieve and mix audio samples with tags corresponding to audio content that the user may find calming or relaxing. Examples of such audio samples might include audio samples tagged as low tempo or low harmonic complexity or audio samples that have been predetermined to be calming or relaxing and have been tagged as such. In some examples, the audio samples can be identified as calming or relaxing based on an automated process that analyzes the temporal and spectral content of the signals. Other examples are possible as well. In any of the examples herein, the generative media module(s) **214** can adjust the characteristics of the generative audio by retrieving and mixing audio samples associated with different metadata tags or other suitable identifiers.

**[0098]** Modifying characteristics of the generative audio can include manipulating one or more of: volume, balance, removing certain instruments or tones, altering a tempo, gain, reverb, spectral equalization, timbre, or sonic texture of the audio, etc. In some examples, the generative audio can be played back differently at different devices, for example emphasizing certain characteristics of the generative audio at the particular playback device that is nearest to the user. For instance, the nearest playback device can emphasize certain instruments, beats, tones, or other characteristics while the remaining playback devices can act as background audio sources.

**[0099]** As described elsewhere herein, the media content module(s) **214** can be configured to produce media intended to direct a user's mood and/or physiological state in a desired direction. In some examples the user's current state (e.g., mood, emotional state, activity level, etc.) is constantly and/or iteratively monitored or measured (e.g., at predetermined intervals) to ensure the user's current state is transitioning toward the desired state or at least not in a direction opposite the desired state. In such examples, the generative audio content can be varied to steer the user's current state towards the desired end state.

**[0100]** In any of the examples herein, the generative media module(s) can use hysteresis to avoid making rapid adjustments to the generative audio that could negatively impact the listening experience. For example, if the generative media module modifies the media based on an input of a user location relative to a playback device, when a user rapidly moves nearer to and farther from a playback device, the playback device could rapidly alter the generative audio in any of the manners described herein. Such rapid adjustments may be unpleasant to the user. In order to reduce these rapid adjustments, the generative media module **214** can be configured to employ hysteresis by delaying the adjustments to the generative audio for a predetermined period of time when the user's movement or other activity triggers an adjustment. For instance, if the playback device detects that the user has moved within the threshold distance of the playback device, then instead of immediately performing

one of the adjustments described above, the playback device can wait a predetermined amount of time (e.g., a few seconds) before making the adjustment. If the user remains within the threshold distance after the predetermined amount of time, then the playback device can proceed to adjust the generative audio. If, however, the user does not remain within the threshold distance after the predetermined amount of time, then the generative media module(s) **214** can refrain from adjusting the generative audio. The generative media module(s) **214** can similarly apply hysteresis to the other generative media adjustments described herein.

**[0101]** FIG. 3 illustrates a flow chart of a process **300** for producing generative audio content using a variety of input parameters. In various examples, one or more of these input parameters can be modified based on user input. For example, an artist may select the various parameters, constraints, or available audio segments shown in FIG. 2, and these selections may in turn determine, at least in part, the final output of generative audio content. As noted previously, such a generative media module may be stored and operated on one or more playback devices for local playback (e.g., via the same playback device and/or via other playback devices communicatively coupled over a local area network). Additionally or alternatively, such a generative media module may be stored and operated on one or more remote computing devices, with the resulting output transmitted over a wide area network to one or more remote devices for playback.

**[0102]** As illustrated, the process begins in block **302** and proceeds to the clock/metronome in block **304**, at which the tempo **306** and time signature **308** inputs are received. The tempo **306** and time signature **308** can be selected by an artist or may be automatically determined or generated using a model. The process continues to block **310** at which a chord change can be triggered, and which receives a chord change frequency parameter **312** as an input. An artist may choose to have a higher chord change frequency in music intended for a higher energy experience (e.g., dance music, uplifting ambient music, etc.). Conversely, a lower chord change frequency may be associated with lower energy output (e.g., calming music).

**[0103]** At block **314**, a chord is selected from the available chord segments **316**. A plurality of chord information parameters **318**, **320**, **322** can also be provided as input to the chord segments **316**. These inputs can be used to determine the particular chord to be played next and output as block **324**. In some examples, the artist can provide information for each chord such as weightings, how often that particular chord should be used, etc.

**[0104]** Next, in block **326**, a chord variation is selected, based at least in part on the harmony complexity parameter which serves as an input. The harmony complexity parameter **328** can be tuned or selected by an artist or may be determined automatically. In general, a higher harmony complexity parameter may be associated with higher energy audio output, and a lower harmony complexity parameter may be associated with lower energy audio output. In some cases, harmony complexity parameters can include inputs such as chord inversions, voicings, and harmony density.

**[0105]** In block **330**, the process gets the root of the chord, and in block **332** selects the bass segment to play from among the available bass segments **334**. These bass seg-

ments are then subject to bus processing **336**, at which equalization, filtering, timing, and other processing can be performed.

[0106] Returning to the chord variation in block **326**, the process separately continues to block **338** to play the harmony selected from among the available harmony segments **340**. This harmony segment is then subject to bus processing **342**. As with the bass bus processing, the harmony segment bus processing **342** can involve equalization, filtering, timing, and other processing can be performed.

[0107] Returning to the selected chord **324**, the process separately continues to filter melody notes in block **344**, which utilizes an input of melody constraints **346**. The output in block **348** is the available melody notes to play. The melody constraints **346** can be provided by the artist, and may, for example, specify which notes to play or not to play, restrict the melody range, or provide other such constraints, which may depend on the particular selected chord **324**.

[0108] In block **350**, the process determines which melody note (from among the available melody notes **348**) to play. This determination can be made automatically based on model values, artist-provided inputs, a randomization effect, or any other suitable input. In the illustrated example, one input is from the trigger melody note block **352**, which in turn is based on the melody density parameter **354**. The artist can provide the melody density parameter **354**, which determines in part how complex and/or high-energy the audio output is. Based on that parameter, a melody note may be triggered more or less frequently and at particular times, using block **352** which is input to block **350** to determine which melody note to play. In various examples, the output of block **350** can be provided as an input to block **350** in the form of a feedback loop, such that the next melody note selected in block **350** depends at least in part on the last selected melody note in block **350**. The melody segment is then selected in block **356** from among the available melody segments **358**, and is then subject to bus processing **360**.

[0109] Returning to the start in block **302**, the process separately proceeds to block **362** to play non-musical content. This may be, for example, nature sounds, spoken-word audio, or other such non-musical content. Various non-musical segments **364** can be stored and available to be played. These non-musical content segments can also be subject to bus processing in block **366**.

[0110] The outputs of these various paths (e.g., the selected bass segment(s), harmony segment(s), melody segment(s), and/or non-musical segment(s)) can each be subject to separate bus processing before being combined at block **368** via mixing and mastering processing. Here the combined levels can be set, various filters can be applied, relative timing can be established, and any other suitable processing steps performed before the generative audio content is output in block **370**. In various examples, some of the paths may be omitted altogether. For example, the generative media module may omit the option of playing back non-musical content alongside the generative musical content. The process **300** shown in FIG. 3 is exemplary only, and one of skill in the art will appreciate that suitable modifications can be made the process **300** shown here, and additionally there are numerous suitable alternative processes that can be used for producing generative media content.

[0111] FIG. 4 is an example architecture for storing and retrieving generative media content. In this example, the

generative media content includes a variety of discrete tracks (each having multiple variations associated with energy level or another parameter), which can be selected and played back in various orders and groupings depending on the particular input parameters.

[0112] As illustrated, the generative media content **404** can be stored as one or more audio files that are associated with global generative media content metadata **402**. Such metadata can include, for example, the global tempo (e.g., beats per minute), global trigger frequency (e.g., how often to check for a change in input parameter(s)), and/or global crossfade duration (e.g., time to fade between the different selected energies).

[0113] Among the generative media content **404** are a plurality of different tracks **406**, **408**, and **410**. In operation, these tracks can be selected and played back in various arrangements (e.g., randomized grouping with some overlap, or played back according to a predetermined sequence, etc.). In some examples, the generative media content **404** including the tracks **406**, **408**, **410** can be stored locally via one or more playback devices, while one or more remote computing devices can periodically transmit updated versions of the tracks, generative media content, and/or global generative media content metadata. In some examples, the remote computing devices can be polled or queried periodically by the playback device(s), and in response to the queries or polls, the remote computing devices can supply updates to the generative media module stored on the local playback device(s).

[0114] For each track, there may be corresponding subsets of that track corresponding to different energy levels. For example, a first energy level (EL) of track 1 at **412**, a second energy level of track 1 at **414**, and an n energy level of track 1 at **416**. Each of these can include both metadata (e.g., metadata **418**, **420**, **422**) and particular media files (e.g., media files **424**, **426**, **428**) corresponding to the particular energy level. In some examples, each track may include a plurality of media files (e.g., media file **424**) arranged in a particular manner, the arrangement and combination of which may be desired by the corresponding metadata (e.g., metadata **418**). The media file can be, for example, in any suitable format that can be played back via the playback device and/or streamed to a playback device for playback. In some examples, one or more of the media files **424**, **426**, **428** can be outputs of the generative model depicted in FIG. 3. The metadata can include, for example, tempo (if different from global tempo), trigger frequency (if different from global trigger frequency), sequence information (e.g., whether to play particular files in order, randomly, or by percent weighting), crossfade duration (if different from global crossfade), spatial information (e.g., for rendering audio content in space using multiple transducers), polyphony information (e.g., allowing multiple audio files to play at once in this segment), and/or level (e.g., level adjustment in dB, or random within a predefined range).

[0115] In operation, one or more input parameters (e.g., number of people present in a room, time of day, etc.) can be used to determine a target energy level. This determination can be made using a playback device and/or one or more remote computing devices. Based on this determination, particular media files can be selected that correspond to the determined energy level. The generative media module can then arrange and play back those selected tracks according to the generative content model. This can involve playing

the selected tracks back in a particular pre-defined order, playing them back in a random or pseudo-random order, or any other suitable approach. The tracks can be played back in a manner that is at least partially overlapping in some examples. It can be useful to vary the amount of overlap between tracks such that casual listeners do not hear a repeating loop of audio content, but instead perceive the generative audio as an endless stream of audio without repetition.

[0116] Although the example shown in FIG. 4 utilizes energy level as a parameter to distinguish different generative audio content, in various examples the particular variations or permutations of generative audio content can vary along other dimensions (e.g., genre, time of day, associated user task, etc.).

#### d. Example Sensor Data Source(s) and Other Input Parameters

[0117] As noted previously, the generative media module (s) 214 can produce generative media based at least in part on input parameters that can include sensor data (e.g., as received from sensor data source(s) 218) and/or other suitable input parameters. With respect to sensor input parameters, the sensor data source(s) 218 can include data from any suitable sensor, wherever located with respect to the generative media group and whatever values are measured thereby. Examples of suitable sensor data includes physiological sensor data such as data obtained from biometric sensors, wearable sensors, etc. Such data can include physiological parameters like heart rate, breathing rate, blood pressure, brainwaves, activity levels, movement, body temperature, etc.

[0118] Suitable sensors include wearable sensors configured to worn or carried by a user, such as a headset, watch, mobile device, brain-machine interface (e.g., Neuralink), headphone, microphone, or other similar device. In some examples, the sensor can be a non-wearable sensor or fixed to a stationary structure. The sensor can provide the sensor data, which can include data corresponding to, for example, brain activity, voice, location, movement, heart rate, pulse, body temperature, and/or perspiration. In some examples, the sensor can correspond to a plurality of sensors. For example, as explained elsewhere herein, the sensor may correspond to a first sensor worn by a first user, a second sensor worn by a second user, and a third sensor not worn by a user (e.g., fixed to a stationary or structure). In such examples, the sensor data can correspond to a plurality of signals received from each of the first, second, and third sensors.

[0119] The sensor can be configured to obtain or generate information generally corresponding to a user's mood or emotional state. In one example, the sensor is a wearable brain sensing headband, which is one of many examples of the sensor described herein. Such a headband can include, for example, an electroencephalography (EEG) headband having a plurality of sensors thereon. In some examples, the headband can correspond to any of the Muse™ headbands (InteraXon; Toronto, Canada). The sensors can be positioned at varying locations around an inner surface of the headband, e.g., to correspond to different brain anatomy (e.g., the frontal, parietal, temporal, and sphenoid bones) of the user. As such, each of the sensors can receive different data from the user. Each of the sensors can correspond to individual channels that can be streamed from the headband

to the system devices 210 and/or 250. Such sensor data can be used to detect a user's mood, for example by classifying the frequencies and intensities of various brainwaves or by performing other analyses. Additional details of using a brain-sensing headband for generative audio content can be found in commonly owned U.S. Application No. 62/706,544, filed Aug. 24, 2020, titled MOOD DETECTION AND/OR INFLUENCE VIA AUDIO PLAYBACK DEVICES, which is hereby incorporated by reference in its entirety.

[0120] In some examples, the sensor data source(s) 218 include data obtained from networked device sensor data (e.g., Internet-of-Things (IoT) sensors such as networked lights, cameras, temperature sensors, thermostats, presence detectors, microphones, etc.). Additionally or alternatively, the sensor data source(s) 218 can include environmental sensors (e.g., measuring or indicating weather, temperature, time/day/week/month, etc.).

[0121] In some examples, the generative media module 214 can utilize input in the form of playback device capabilities (e.g., number and type of transducers, output power, other system architecture), device location (e.g., either location relative to other playback devices, relative to one or more users). Additional examples of creating and modifying generative audio as a result of user and device location are described in more detail in commonly owned U.S. Application No. 62/956,771, filed Jan. 3, 2020, titled GENERATIVE MUSIC BASED ON USER LOCATION, which is hereby incorporated by reference in its entirety. Additional inputs can include a device state of one or more devices within the group, such as a thermal state (e.g., if a particular device is in danger of overheating, the generative content can be modified to reduce temperature), battery level (e.g., bass output can be reduced in a portable playback device with low battery levels), and bonding state (e.g., whether a particular playback device is configured as part of a stereo pair, bonded with a sub, or as part of a home theatre arrangement, etc.). Any other suitable device characteristic or state may similarly be used as an input for production of generative media content.

[0122] Another example input parameter includes user presence—for example when a new user enters a space playing back generative audio, the user's presence can be detected (e.g., via proximity sensors, a beacon, etc.) and the generative audio can be modified as a response. This modification can be based on number of users (e.g., with ambient, meditative audio for 1 user, relaxing music for 2-4 users, and party or dance music for greater than 4 users present). The modification can also be based on the identity of the user(s) present (e.g., a user profile based on the user's characteristics, listening history, or other such indicia).

[0123] In one example, a user can wear a biometric device that can measure various biometric parameters, such as heart rate or blood pressure, of the user and report those parameters to the devices 210 and/or 250. The generative media modules 214 of these devices 210 and/or 250 can use these parameters to further adapt the generative audio, such as by increasing the tempo of the music in response to detecting a high heart rate (as this may indicate that the user is engaging in a high motion activity) or decreasing the tempo of the music in response to detecting a high blood pressure (as this may indicate that the user is stressed and could benefit from calming music).

[0124] In yet another example, one or more microphones of a playback device (e.g., microphones 115 of FIG. 1F) can

detect a user's voice. The captured voice data can then be processed to determine, for example, a user's mood, age, or gender, to identify a particular user from among several users within a household, or any other such input parameter. Other examples are possible as well.

#### e. Example Coordination Among Group Members

[0125] FIG. 5 is a functional block diagram illustrating data exchange in a system for playback of generative media content. For purposes of explanation, the system 500 shown in FIG. 5 includes interactions between a coordinator device 210 and a member device 250b. However, the interactions and processes described herein can be applied to interactions involving a plurality of additional coordinator devices 210 and/or member devices 250. As shown in FIG. 5, the coordinator device 210 includes a generative media module 214a that receives inputs including input parameters 502 (e.g., sensor data, media content, model parameters for the generative media module 214a, or other such input) as well as clock and/or timing data 504. In various examples, the clock and/or timing data 504 can include synchronization signals to synchronize playback and/or to synchronize generative media being produced by various devices within the group. In some examples, the clock and/or timing data 504 can be provided by an internal clock, processor, or other such component housed within the coordinator device 210 itself. In some examples, the clock and/or timing data 504 can be received via a network interface from remote computing devices.

[0126] Based on these inputs, the generative media module 214a can output generative media content 404a. Optionally, the output generative media content 404a can itself serve as an input to the generative media module 214a in the form of a feedback loop. For example, the generative media module 214a can produce subsequent content (e.g., audio frames) using a model or algorithm that depends at least in part on the previously generated content.

[0127] In the illustrated example, the member device 250b likewise includes a generative media module 214b, which can be substantially the same as the generative media module 214a of the coordinator device 210, or may differ in one or more aspects. The generative media module 214b can likewise receive input parameter(s) 502 and clock and/or timing data 504. These inputs can be received from the coordinator device 210, from other member devices, from other devices on a local network (e.g., a locally networked smart thermostat supplying temperature data), and/or from one or more remote computing devices (e.g., a cloud server providing clock and/or timing data 504, or weather data, or any other such input). Based on these inputs, the generative media module 214b can output generative media content 404b. This produced generative media content 404b can optionally be fed back into the generative media module 214b as part of a feedback loop. In some examples, the generative media content 404b can include or consist of generative media content 404a (produced via the coordinator device 210) which has been transmitted over a network to the member device 250b. In other cases, the generative media content 404b can be produced independently and separately of the generative media content 404a produced via the coordinator device 210.

[0128] The generative media content 404a and 404b can then be played back, either via the devices 210 and 250b themselves, and/or as played back by other devices within

the group. In various examples, the generative media content 404a and 404b can be configured to be played back concurrently and/or synchronously. In some instances, the generative media content 404a and 404b can be substantially identical or similar to one another, with each generative media module 214 utilizing the same or similar algorithms and the same or similar inputs. In other instances, the generative media content 404a and 404b can differ from one another while still being configured for synchronous or concurrent playback.

#### f. Example Generative Media Using Distributed Architectures

[0129] As noted previously, generation of media content can be computationally intensive, and in some cases may be impractical to perform wholly on local playback devices alone. In some examples, a generative media module of a local playback device can request generative media content from a generative media module stored on one or more remote computing devices (e.g., cloud servers). The request can include or be based on particular input parameters (e.g., sensor data, user inputs, contextual information, etc.). In response to the request, the remote generative media module can stream particular generative media content to the local device for playback. The particular generative media content provided to the local playback device can vary over time, with the variation depending on the particular input parameters, the configuration of the generative media module, or other such parameter. Additionally or alternatively, the playback device can store discrete tracks for playback (e.g., with different variations of tracks associated with different energy levels, as depicted in FIG. 4). The remote computing device(s) may then periodically provide new files for updated tracks to the local playback device for playback, or alternatively may provide an update to the generative media module that determines when and how to play back the particular files that are stored locally on the playback device.

[0130] In this manner, the tasks required to produce and play back generative audio are distributed among one or more remote computing device(s) and one or more local playback devices. By performing at least some of the computationally intensive tasks associated with generating novel media content to the remote computing devices, and optionally by reducing the need for real-time computation, overall efficiency can be improved. By generating, via remote computing devices, a discrete number of alternative tracks or track variation according to a particular media content model ahead of playback, the local playback device may request and receive particular variations based on real-time or near-real-time input parameters (e.g., sensor data). For example, the remote computing devices can generate different versions of the media content, and the playback device can request particular versions in real-time based on input parameters. The result is playback of suitable generative media content based on real-time or near-real-time input parameters (e.g., sensor data) without requiring de novo generation of such media content to be performed in real time.

[0131] FIG. 6 is a schematic diagram of an example distributed generative media playback system 600. As illustrated, an artist 602 can supply a plurality of media segments 604 and one or more generative content models 606 to a generative media module 214 stored via one or more remote computing devices. The media segments can correspond to,

for example, particular audio segments or seeds (e.g., individual notes or chords, short tracks of  $n$  bars, non-musical content, etc.). In some examples, the generative content models **606** can also be supplied by the artist **602**. This can include providing the entire model, or the artist **602** may provide inputs to the model **606**, for example by varying or tuning certain aspects (e.g., tempo, melody constraints, harmony complexity parameter, chord change density parameter, etc.).

[0132] The generative media module **214** can receive both the media segments **604** and one or more input parameters **502** (as described elsewhere herein). Based on these inputs, the generative media module **214** can output generative media. As shown in FIG. 6, the artist **602** can optionally audition the generative media module **214**, for example by receiving exemplary outputs based on the inputs provided by the artist **602** (e.g., the media segments **604** and/or generative content model(s) **606**). In some cases, the audition can play back to the artist **602** variations of the generative media content depending on a variety of different input parameters (e.g., with one version corresponding to a high energy level intended to produce an exciting or uplifting effect, another version corresponding to a low energy level intended to produce a calming effect, etc.). Based on the outputs via this audition step, the artist **602** may dynamically update the media segments **604** and/or settings of the generative content model(s) **606** until the desired outputs are achieved.

[0133] In the illustrated example, there can be an iteration at block **608** every  $n$  hours (or minutes, days, etc.) at which the generative media module **214** can produce a plurality of different versions of the generative media content. In the illustrated example, there are three versions: version A in block **610**, version B in block **612**, and version C in block **614**. These outputs are then stored (e.g., via the remote computing device(s)) as generative media content **616**. A particular one of the versions (version C as block **618** in this example) can be transmitted (e.g., streamed) to the local playback device **250** for playback. In some examples, the particular versions can correspond to the tracks **406**, **408**, and **410** shown in FIG. 4.

[0134] Although three versions are shown here by way of example, in practice there may be many more versions of generative media content produced via the remote computing devices. The versions can vary along a number of different dimensions, such as being suitable for different energy levels, suitable for different intended tasks or activities (e.g., studying versus dancing), suitable for different time of day, or any other appropriate variations.

[0135] In the illustrated example, the playback device **250** can periodically request a particular version of the generative media content from the remote computing device(s). Such requests can be based on, for example, user inputs (e.g., user selection via a controller device), sensor data (e.g., number of people present in a room, background noise levels, etc.), or other suitable input parameter. As illustrated, the input parameter(s) **502** can optionally be provided to (or detected by) the playback device **250**. Additionally or alternatively, the input parameter(s) **502** can be provided to (or detected by) the remote computing device(s) **106**. In some examples, the playback device **250** transmits the input parameters to the remote computing device(s) **106**, which in turn provide a suitable version to the playback device **250**, without the playback device **250** specifically requesting a particular version.

#### g. Example Methods for Generation of Digital Content Based on Blockchain Data

[0136] As noted previously, in some instances, a system for generating and playing back generative media content can interact with blockchain data (or data stored via other distributed ledger technology). For example, as shown in FIG. 6, a blockchain layer **620** can be utilized to provide data as an input to other components of the generative media playback system **600**, such as the playback device **250**, the input parameters **502**, the media segments **604**, the generative content model(s) **606**, and/or the generative media module **214**. In various instances, the blockchain layer **620** can store data that can be used as one or more input parameters **502**, data that contains or can be used to obtain or affect particular media segments **604**, data that contains or can be used to obtain or affect particular generative content model(s) **606**, and/or data that contains or can be used to obtain or affect particular generative media modules **214**. Additionally, some or all of these components may communicate with the blockchain layer **620** to write data to the blockchain, record transactions, or otherwise interact with the blockchain layer **620**. For example, the playback device **250** may record transactions that reflect the playback of particular tracks on the blockchain layer **620**. Data stored via the blockchain layer **620** can include particular input parameters **502**, or data stored via the blockchain layer **620** can be used to generate suitable input parameters **502**. Similarly, particular media segments **604**, generative content models **606**, generative media modules **214**, input parameters **502**, or other suitable data can be written to the blockchain layer **620** to create an immutable record of such content, transactions, or other data. Additional details regarding utilization of blockchain technology (or other suitable distributed ledger technology) in the creation and playback of generative media content are described in more detail below.

[0137] FIG. 7 is a schematic diagram of another example distributed generative media playback system **700**. As shown, the system **700** includes or communicates with a blockchain layer **620**, for example to obtain, generate, or store input parameters **502**, generative content models **606**, or other data or parameters used in the creation and playback of generative media content.

[0138] Examples of such blockchain layers **620** include public distributed ledgers such as Ethereum, Bitcoin, Solana, Avalanche, Polygon, and many others. Although a blockchain layer **620** is illustrated, in various examples any suitable distributed ledger technology may be used, including private or semi-private blockchains, as well as non-blockchain implementations such as directed acyclic graphs (DAGs) (e.g., Nano, IOTA, etc.). In various examples, participants using a blockchain layer **620** can transact with one another in a peer-to-peer manner, and operation of the blockchain layer **620** can be distributed such that no one central entity controls operation of the network. Such distributed ledgers can be used to track the creation, exchange, and redemption of certain real-world assets, such as currency. This approach enables robust auditing of asset transactions due to the practical immutability of data stored on a blockchain. Currency is only one of various assets that may be desirable to track on a distributed ledger. Other types of assets may differ from currency with respect to one or more behaviors that govern asset creation, exchange, and/or redemption. Moreover, different blockchain architectures

may differ with respect to policies and protocols, and further with respect to the tools used to program asset behaviors.

**[0139]** In general, a distributed ledger in which each unit of an asset is represented by some form of digital token can be programmed to endow that token with a set of behaviors appropriate for the asset it represents. By way of example, “fungible” behavior enables an asset to be exchanged with other assets of the same class. Every unit of a given denomination of currency (e.g., a dollar) is fungible, because it has the same value as every other unit of the same denomination. A property title, by contrast, is “non-fungible” because its value depends on the size, location, and other aspects of the specified property. For each asset represented as a token, the appropriate fungible or non-fungible behavior is programmed into that token’s class in the virtual ledger that tracks the asset.

**[0140]** According to some embodiments, tokens transacted via the blockchain layer **620** can be non-fungible. Such non-fungible tokens (NFTs) can be unique and non-interchangeable for any other token. The NFT may comprise and/or be associated with a unique digital artwork and/or music, domain name, a digital collectible (e.g., CryptoKitties, memes), an event ticket, part of a virtual world, a digital object used in games, an avatar or character, an item having utility (e.g., a specific function such as providing voting or governance rights), etc. In various examples, the NFTs may themselves include the relevant data (e.g., the raw audio data for a music NFT may be stored on-chain), or the NFT may include a pointer (e.g., URL or URI) that directs to data stored elsewhere (e.g., audio data stored on a server maintained by the issuer of the music NFT).

**[0141]** Such tokens, whether fungible or non-fungible, may be stored by users via a digital wallet. A digital wallet can be a device, physical medium, program, or service that stores the public and/or private keys for blockchain transactions. In some instances, a digital wallet can store multiple public/private key pairs for various different blockchains, such that a user can store assets associated with different blockchains in a single wallet. Examples include MetaMask, Phantom, Coinbase Wallet, Ledger Nano, etc. In operation, a user can sign blockchain transactions via the wallet using the appropriate private key (or authorizing the wallet to sign the transaction with the private key). The transaction will then be confirmed if the transaction signature is valid, and then added to a corresponding block in the blockchain. In some examples, a wallet identifier may itself be used as an input parameter to a generative model, independent of any tokens held in the particular wallet.

**[0142]** In various implementations, the blockchain layer **620** can be configured to execute transactions automatically under one or more conditions. Such self-executing transactions can be referred to as “smart contracts.” A smart contract may comprise computer code stored on the blockchain that is configured to execute only in specified circumstances and in specified manners. For example, a smart contract can be configured such that a particular transaction executes when a threshold is exceeded, at a certain time, based on one or more other transactions, or any other suitable criteria. In some examples, a generative media module **214** and/or a generative content model **606** can be implemented in the form of a smart contract, such that interacting with the smart contract through the blockchain layer **620** causes the smart contract to output generative media content, a generative content model, or data or

instructions that can be used to produce such a generative media content or a generative content model.

**[0143]** One organizational structure unique to blockchains is a decentralized autonomous organization (DAO). A DAO is typically a community-led entity with no central authority. Such DAOs may be fully autonomous and transparent, with smart contracts providing the foundational rules and executing the agreed-upon decisions. Community votes may be cast via token holders using on-chain transactions. Based on the outcome of particular votes, smart contracts can execute certain transactions or other code to implement the decisions of the DAO members. Typically, DAOs issue tokens to users, either in exchange for currency investment or donations, or without compensation (e.g., via an “air drop”). Token holders typically then retain certain voting rights, which may be proportional to their holdings. In some instances, token holders also receive financial proceeds, such as a share of transaction fees collected by the DAO.

**[0144]** In the example system **700** shown in FIG. 7, the generative media module **214** can receive a number of different inputs and output one or more generative content versions **610** in response. These content versions **610** are stored in generative media content store **616**, from which a particular selected generative content version **618** can be chosen for playback via playback device **250** or other output device (e.g., a light component of the generative media content version **618** can cause the illumination device **702** to output light in accordance with the generative media content version **618**, such as having a particular hue, color temperature, brightness, on/off or other pattern, etc.). The inputs to the generative media module **214** include the generative content model **606** and input parameter(s) **502**, similar to the approach described previously with respect to FIG. 6. As also noted previously, in some instances the generative content model **606** can be stored via the blockchain layer **620** or can be obtained from data stored via the blockchain layer **620**. Similarly, one or more of the input parameter(s) **502** can include or be based on data stored via the blockchain layer **620**. For example, blockchain data can be used in the generative media module **214** to produce suitable output, such as the “sonification” of data streams (e.g., a real-time feed of cryptocurrency price values or other data may be turned into a corresponding sound output). In some instances, the blockchain data can include data provided by one or more “oracles,” which are typically third-party services that provide smart contracts with external information (e.g., price feeds, weather data, election results, etc.). In some examples, the generative media module **606** and/or the generative media content **616** may be stored locally via the playback device **250**, in which case the input parameters **502** may be streamed to the playback device **250** for use in generating new versions **610** of the generative media content **616**.

**[0145]** Additionally or alternatively, the generative media module **214** can receive, as an input, the output of one or more smart contracts **706**. In some instances, the generative media module **214** may itself take the form of a smart contract, such that the program code is stored on the blockchain and runs automatically under certain conditions (e.g., a user **708** interacts with the smart contract **706** and in response particular generative media content is transmitted to a specified destination). In some examples, the generative media module **214** may run locally or via remote servers (rather than as a smart contract on a blockchain), but may

communicate with the smart contract **706** either to receive input parameters **502** therefrom, or to provide a suitable output to the smart contract **706**. For example, particular generative media content output by the generative media module **214** can be used to generate one or more NFTs via the smart contract **706**. In some implementations, each particular version of the generative content produced by the generative media module **214** can have a corresponding NFT, such that each NFT produced by the smart contract **706** based on input from the generative media module **214** is unique. This is illustrated graphically in FIG. 7 with a plurality of discrete NFTs **710a-f** labeled NFT1-NFTn that may be produced via smart contract **706**. These NFTs **710** can additionally be provided as inputs to the generative media module **214**. For example, the generative media module **214** may produce dynamically distinct content based at least in part on the particular NFTs **710** with which it interacts. Additionally or alternatively, a user **708** may only be able to access the particular generative media module **214** if the user holds the appropriate NFT **710** in her digital wallet.

[0146] In some examples, one or more of the NFTs **710** are existing NFTs owned by individual third-party persons or entities (e.g., persons or entities not associated with the user **708**) and accessible, perhaps on a temporary basis, by the generative media module **214**. In certain examples, one or more of the NFTs **710** do not comprise audio data, but instead comprise other data types (e.g., video, image, or other data) that can be “sonified” or otherwise translated by the generative media module **214** (or another suitable component) into a form that can be used to generate media content.

[0147] In the illustrated example, the smart contract **706** can also output NFT **712**, labeled NFT0, which may be held by the user **708**. Additionally, this NFT **712** can interact with, or be generated via, the artist DAO **714**, which in turn may communicate with one or more smart contracts **706**. As noted previously, DAOs are typically community-led organizations in which members hold tokens (e.g., NFT **712**) designating membership, providing voting or other governance rights, and optionally entitling the token holders to economic benefits such as proceeds from future DAO revenue. In some examples, the artist DAO **714** may hold certain music royalties which, when collected, can be distributed in part to holders of the appropriate NFTs or other tokens (e.g., user **708** may receive economic benefits from the artist DAO **714** based at least in part on the user's ownership of the NFT **712**).

[0148] Optionally, data corresponding to the NFT **712** can be stored or embedded via physical media. For example, as shown in FIG. 7, data corresponding to the NFT **712** can be embedded on a vinyl record **716** (e.g., via a unique QR code, a code embedded in grooves of the vinyl record **716**), or any other suitable technique for storing data via the physical media (e.g., NFC or other RF tag). Although a vinyl record **716** is illustrated, in various examples the physical substrate can take any number of forms, for example a playback device, a physical card or ticket, a poster, etc.

[0149] The inclusion of blockchain layer **620**, smart contract(s) **706**, DAO **714**, and/or NFTs **710** and **712** can provide several benefits for the generative media playback system **700**. For example, by associating particular NFTs with generative media content (e.g., soundscapes), the user **708** can obtain a personalized experience that is also poten-

tially transferrable via the distributed peer-to-peer transaction mechanism of the blockchain layer **620**. One problem with this approach, however, is that the data contained in the NFT is typically static, which is antithetical to the dynamic, contextually aware nature of a generative soundscape or other generative media content. Another problem associated with NFTs is link rot, in which the data in the NFT is rendered obsolete because the locators in the NFT no longer refer to the artwork associated with the NFT. One way to mitigate the possibility of link rot is to store generative media content or a generative media engine in a blockchain layer **620**. Additionally, in some instances the artwork may itself be embedded into the NFT (e.g., the artwork data is stored on-chain rather than on a separate server).

[0150] In some instances, the NFTs **710** can include the particular seeds used by the generative media module **214**. Examples of such seeds include media segments **604** of FIG. 6, tracks, energy levels, or metadata of FIG. 4, or any of the various components shown in FIG. 3). Optionally, the properties of the particular NFT (at least with respect to its use by the generative media module **14**) may depend on its transaction history. For example, depending on when an NFT was last transacted, or on how many transactions it has been subjected to, the particular seed associated with that NFT may vary dynamically. Additionally or alternatively, different combinations of NFTs **710** connected to the generative media module **214** can result in different content versions **610** being produced, such that the particular generative media output will depend on which NFTs **710** are used as inputs.

[0151] In some examples, additional data associated with the generative media playback system **700** can be stored via the blockchain layer **620**. For example, the listening history of the user **708** can be stored to the blockchain layer **620** to provide an immutable record of listening history. This can include listening history of generative media content or non-generative content (e.g., standard pre-recorded audio tracks or other content). In some instances, “followers” of a particular user **708** can subscribe to that user's listening history by accessing the data stored via the blockchain layer **620**. As blockchains are typically permissionless and transparent, followers may be free to access the listening history (or other content data associated with a particular network address). In yet another example, a follower may subscribe to the generative media content of a particular user **708**, such that while the generative media content for the user **708** is created dynamically based on the various inputs, this same media content can be available for other followers to enjoy.

[0152] Consider, for example, an artist who wants to create a particular soundscape using a generative media module **214**. The artist's fans may listen to the soundscape in real-time or nearly real-time via blockchain data. In some instances, the followers may utilize their own local generative media modules **214** (or those running on other devices) that then use inputs, pointers, or other data from the artist's generative media module **214** to produce corresponding generative media content. In at least some examples, such local generative media modules **214** may also utilize additional local inputs (e.g., particular playback device characteristics, local sensor data, etc.) such that the artist's generative media content is merged with that produced by the user's own local generative media module **214** to produce somewhat altered generative media content that nonetheless reflects the artist's intention. Optionally, an artist's followers

may be required to hold a particular NFT or other token to access that artist's generative media content. In yet another example, a particular curated playlist or radio station may be accessible only to users who hold a particular NFT or token.

#### h. Example Methods for Generation and Playback of Generative Audio

[0153] FIGS. 8-13 are flow diagrams of example methods for playing back generative audio content via multiple discrete playback devices. The methods **800**, **900**, **1000**, **1100**, **1200**, **1300** can be implemented by any of the devices or systems described herein, or any other devices or systems now known or later developed.

[0154] Various examples of the methods **800**, **900**, **1000**, **1100**, **1200**, **1300** include one or more operations, functions, or actions illustrated by blocks. Although the blocks are illustrated in sequential order, these blocks may also be performed in parallel, and/or in a different order than the order disclosed and described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon a desired implementation.

[0155] In addition, for the methods **800**, **900**, **1000**, **1100**, **1200**, **1300**, and for other processes and methods disclosed herein, the flowcharts show functionality and operation of possible implementations of some examples. In this regard, each block may represent a module, a segment, or a portion of program code, which includes one or more instructions executable by one or more processors for implementing specific logical functions or steps in the process. The program code may be stored on any type of computer readable medium, for example, such as a storage device including a disk or hard drive. The computer readable medium may include non-transitory computer readable media, for example, such as tangible, non-transitory computer-readable media that stores data for short periods of time like register memory, processor cache, and Random-Access Memory (RAM). The computer readable medium may also include non-transitory media, such as secondary or persistent long-term storage, like read only memory (ROM), optical or magnetic disks, compact disc read only memory (CD-ROM), for example. The computer readable media may also be any other volatile or non-volatile storage systems. The computer readable medium may be considered a computer readable storage medium, for example, or a tangible storage device. In addition, for the methods and for other processes and methods disclosed herein, each block in FIGS. 8-13 may represent circuitry that is wired to perform the specific logical functions in the process.

[0156] With reference to FIG. 8, the method **800** begins at block **802**, which involves receiving a command to play back generative media content via a group or bonded zone of playback devices. Such a command can be received via, for example a control device **130** or other suitable user input.

[0157] At block **804**, the method **800** involves a group coordinator device providing timing information to generative group member devices. The timing information can include contextual timing data (e.g., time data associated with sensor input or other user input), generative media playback timing data (e.g., time stamps and synchronization data to facilitate synchronous playback of generative media), and/or media content stream timing data based on a common clock.

[0158] At block **806**, the method optionally includes determining generative media content model(s) to be used to produce generative media. Such models can be implemented in, for example, the media content module(s) **214** described above with respect to FIGS. 2-6. In some examples, each of the member devices can utilize the same or substantially the same generative media content models, while in other cases some or all of the member devices can utilize generative media content models that differ from one another. For example, a first generative media content model can produce rhythmic beats, while a second generative media content model can produce ambient nature sounds. When played back concurrently, the generative audio produced by these different generative media content models can produce a pleasing listener experience for the users. In some examples, the selection of a particular generative media content model can itself be based on one more input parameters, such as device capabilities, device location, number of users present, user sensor data, etc.

[0159] In block **808**, the method **800** includes the coordinator device and member device(s) receiving contextual and/or other input data. For example, the input data can include sensor data, user inputs, context data, or any other relevant data that can be utilized as an input for a generative media content model.

[0160] The method **800** continues in block **810** with the coordinator and member devices generating and playing back generative media content in synchrony.

[0161] FIG. 9 illustrates another method **900** for playing back generative audio content via multiple playback devices. The method **900** begins in block **902** with receiving, at a group coordinator device, one or more input parameters. As noted previously, the input parameters can include sensor data, user input, contextual data, or any other input that can be used by a generative media module to produce generative audio for playback.

[0162] In block **904**, the coordinator device transmits the input parameters to one or more discrete playback devices having generative media modules thereon. For example, the coordinator device can obtain sensor data and other input parameters and transmit these to a plurality of discrete playback devices within an environment or even distributed among multiple environments. In some examples, these input parameters can include features of the generative content model(s) themselves, for example providing instructions to update the generative media modules stored by locally by one or more of the discrete playback devices.

[0163] In block **906**, the method involves transmitting timing data from the coordinator device to the playback devices. The timing data can include, for example clock data or other synchronization signals configured to facilitate coordination of the production of generative media content as well as synchronous playback of that generative media content via the discrete playback devices.

[0164] The method **900** continues in block **908** with concurrently playing back generative media content via the playback devices based at least in part on the input parameters. As discussed previously, the various playback devices may play back the same generative audio or each may play back distinct generative audio that, when played back synchronously, produces a desired psychoacoustic effect for the users present.

[0165] In the example of FIG. 9, the generative media content can be produced locally by discrete playback

devices, which each play back creating their own generative audio content in parallel with one another. In an alternative method **1000** shown in FIG. **10**, the generative media content is produced at the coordinator device, which then transmits the generative media content along with timing data to the discrete playback devices for synchronous playback.

[0166] In block **1002**, the method **1000** involves receiving, at a group coordinator device, one or more input parameters. Examples of the input parameters are described elsewhere herein, and include sensor data, user input, contextual data, or any other input that can be used by a generative media module to produce generative audio for playback.

[0167] In block **1004**, the coordinator device generates first and second generative media streams based at least in part on the input parameters, and in block **1006** the first and second media streams are transmitted to first and second discrete playback devices, respectively. For example, the coordinator device can generate two streams that form different channels of generative audio, for example with a left channel to be played back by a first playback device and a corresponding right channel to be played back by a second playback device. Additionally or alternatively, the two streams can be distinct audio tracks that nonetheless can be played back synchronously, such as rhythmic beats in one stream and ambient nature sounds in the other stream. Multiple other variations are possible. Although this example describes two streams for two playback devices, in various other examples there may be one stream or more than two streams that can be provided to any number of playback devices for synchronous playback. In at least some examples, one or more of the playback devices can be positioned in different environments far apart from one another (e.g., in different households, different cities, etc.).

[0168] In block **1008**, the first playback device plays back the first generative media stream and the second playback device concurrently plays back the second generative media stream. In some examples, this concurrent playback can be facilitated by use of timing data received from the coordinator device.

[0169] FIGS. **11** illustrates another example method **1100** for generation and playback of generative media content. As described above, it can be beneficial to perform at least a portion of the processing required to produce generative media content using one or more remote computing devices (e.g., cloud-based servers) so as to reduce the computational demands placed on local playback devices and/or to perform operations that would not be feasible using the components of the local playback devices. The method **1100** begins in block **1102** with receiving, at a playback device, one or more input parameters. As noted previously, the input parameters can include sensor data, user input, contextual data, or any other input that can be used by a generative media module to produce generative audio for playback.

[0170] In block **1104**, the method **1100** involves accessing a library that includes a plurality of pre-existing media segments. For example, a plurality of discrete media segments (e.g., audio tracks) can be stored on the playback device, and can be arranged and/or mixed for playback according to a generative content model. Additionally or alternatively, the library can be stored on one or more remote computing devices, with individual media segments being transmitted from the remote computing device(s) to the playback device for playback.

[0171] The method **1100** continues in block **1106** with generating media content by arranging a selection of the pre-existing media segments from the library for playback according to a generative media content model and based, at least in part, on the input parameters. As described elsewhere herein, a generative media content model can receive the one or more input parameters as an input. Based on the input, and using the generative media content model, particular generative media content can be output. Among examples, the generative media content can include an arrangement of the pre-existing media segments, for example arranging them in a particular order, with or without overlap between the particular media segments, and/or with additional processing or mixing steps performed to produce the desired output.

[0172] In block **1108**, the playback device plays back the generated media content. In various examples, this playback can be performed concurrently and/or synchronously with additional playback devices.

[0173] FIG. **12** illustrates another example method **1200** for generation and playback of generative media content. As described above, it can be beneficial to incorporate or rely upon blockchain data to produce generative media content. The method **1200** begins in block **1202** with accessing, via a playback device, blockchain data stored via a distributed ledger. The distributed ledger can be a public blockchain such as Ethereum, Bitcoin, Solana, etc., or may optionally be a private or semi-private blockchain or non-blockchain ledger. The blockchain data can include one or more pre-existing media segments or other seeds to be used in producing generative media content. In some examples, such data is stored directly on the blockchain itself, while in other instances the blockchain can store a pointer (e.g., a URL or URI) that directs to the stored location of the media segments or other seed data. Optionally, the blockchain data takes the form of one or more non-fungible tokens (NFTs).

[0174] The method **1200** continues in block **1204** with generating, via the playback device, media content based at least in part on the blockchain data. In some examples, this generation can include accessing a library of pre-existing media segments that is stored on the playback device or other suitable storage location (e.g., remote server, other devices on a local network, etc.). In some instances, these media segments can be retrieved from the blockchain or other remote location and stored via the playback device. The playback device may then arrange a selection of pre-existing media segments from the library for playback according to a generative media content model. This selection can be based at least in part on the blockchain data. For example, as described elsewhere herein, the generative media content model may communicate with a smart contract or a decentralized autonomous organization (DAO) in a manner that affects the particular generative media content that is output by the model. In some instances, particular NFTs or other tokens can affect the output of the generative media content model. Such NFTs or other tokens may, in some instances, be used in combination, such that the particular combination of NFTs or other tokens produces a unique output via the generative media content model. In at least some instances, two or more blockchains can be utilized concurrently in this manner (e.g., the generative media content model can vary the output based on a user holding both a first NFT on the Solana network and a second

NFT on the Ethereum network). In block **1206**, the method **1200** involves playing back, via the playback device, the generated media content.

**[0175]** FIG. **13** illustrates another example method **1300** for generation and playback of generative media content. As noted above, smart contracts or other self-executing code can be used to produce, store, and playback generative media content. The method **1300** begins in block **1302** with transmitting, via a playback device, data associated with a first token over a network to a network address of a distributed ledger. The address can be associated with a generative media smart contract configured to produce a generative media content model. In some examples, the first token can be an NFT, and optionally a plurality of such tokens can be transmitted to the smart contract address.

**[0176]** In block **1304**, the method **1300** involves receiving, via the playback device, the generative media content model from the network address associated with the generative media smart contract. For example, when the smart contract is executed, it can produce a particular generative media content model that is based at least in part on the data associated with the first token. This generative media content model can then be provided to a user for generation of novel media content that is tailored based on the first token data. In addition to the first token data, the smart contract may also produce different outputs based on other input parameters as described elsewhere (e.g., sensor data, playback device characteristic data, playback device state, user listening history data, etc.), provided that such data is provided to the smart contract address. Additionally or alternatively, the generative media content model provided to the user can output different media content based on one or more of these other input parameters.

**[0177]** Next, in block **1306**, the playback device generates media content based at least in part on the generative media content model. In some examples, this generation can include accessing a library of pre-existing media segments that is stored on the playback device or other suitable storage location (e.g., remote server, other devices on a local network, etc.). The playback device may then arrange a selection of pre-existing media segments from the library for playback according to the generative media content model. In block **1308**, the method **1300** involves playing back, via the playback device, the generated media content.

**[0178]** Various examples of generative media playback are described herein. One of ordinary skill in the art will understand that a wide variety of different generative media modules, algorithms, inputs, sensor data, and playback device configurations are contemplated and may be used in accordance with the present technology.

#### IV. Conclusion

**[0179]** The above discussions relating to playback devices, controller devices, playback zone configurations, and media content sources provide only some examples of operating environments within which functions and methods described below may be implemented. Other operating environments and configurations of media playback systems, playback devices, and network devices not explicitly described herein may also be applicable and suitable for implementation of the functions and methods.

**[0180]** The description above discloses, among other things, various example systems, methods, apparatus, and articles of manufacture including, among other components,

firmware and/or software executed on hardware. It is understood that such examples are merely illustrative and should not be considered as limiting. For example, it is contemplated that any or all of the firmware, hardware, and/or software aspects or components can be embodied exclusively in hardware, exclusively in software, exclusively in firmware, or in any combination of hardware, software, and/or firmware. Accordingly, the examples provided are not the only ways) to implement such systems, methods, apparatus, and/or articles of manufacture.

**[0181]** Additionally, references herein to “example” means that a particular feature, structure, or characteristic described in connection with the example can be included in at least one example or embodiment of an invention. The appearances of this phrase in various places in the specification are not necessarily all referring to the same example, nor are separate or alternative examples mutually exclusive of other examples. As such, the examples described herein, explicitly and implicitly understood by one skilled in the art, can be combined with other examples.

**[0182]** The specification is presented largely in terms of illustrative environments, systems, procedures, steps, logic blocks, processing, and other symbolic representations that directly or indirectly resemble the operations of data processing devices coupled to networks. These process descriptions and representations are typically used by those skilled in the art to most effectively convey the substance of their work to others skilled in the art. Numerous specific details are set forth to provide a thorough understanding of the present disclosure. However, it is understood to those skilled in the art that certain examples of the present technology can be practiced without certain, specific details. In other instances, well known methods, procedures, components, and circuitry have not been described in detail to avoid unnecessarily obscuring aspects of the examples. Accordingly, the scope of the present disclosure is defined by the appended claims rather than the foregoing description of examples.

**[0183]** When any of the appended claims are read to cover a purely software and/or firmware implementation, at least one of the elements in at least one example is hereby expressly defined to include a tangible, non-transitory medium such as a memory, DVD, CD, Blu-ray, and so on, storing the software and/or firmware.

**[0184]** The disclosed technology is illustrated, for example, according to various examples described below. Various examples of examples of the disclosed technology are described as numbered examples (1, 2, 3, etc.) for convenience. These are provided as examples and do not limit the disclosed technology. It is noted that any of the dependent examples may be combined in any combination, and placed into a respective independent example. The other examples can be presented in a similar manner.

**[0185]** Example 1: A method comprising: receiving, at a coordinator device, input parameters; transmitting the input parameters from the coordinator device to a plurality of playback devices each having a generative media module therein; transmitting timing data, from the coordinator device to the plurality of playback devices, such that the playback devices concurrently play back generative media content based at least in part on the input parameters.

**[0186]** Example 2: The method of any one of the Examples herein, wherein first and second playback devices

play back different generative audio content, each based at least in part on the input parameters.

**[0187]** Example 3: The method of any one of the Examples herein, wherein the input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors, microphones); environmental data (e.g., weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity), user mood data).

**[0188]** Example 4: The method of any one of the Examples herein, wherein the timing data comprises at least one of: clock data or one or more synchronization signals.

**[0189]** Example 5: The method of any one of the Examples herein, further comprising transmitting a signal from the coordinator device to at least one of the plurality of playback devices that causes the generative media module of the playback device to be modified.

**[0190]** Example 6: The method of any one of the Examples herein, wherein the generative media content comprises at least one of: generative audio content or generative visual content.

**[0191]** Example 7: The method of any one of the Examples herein, wherein the generative media modules comprise algorithms that automatically generate novel media output based on inputs that include at least the input parameters.

**[0192]** Example 8: A device comprising: a network interface; one or more processors; and a tangible, non-transitory computer-readable medium storing instructions that, when executed by the one or more processors, cause the device to perform operations comprising: receiving, via the network interface, input parameters; transmitting, via the network interface, the input parameters to a plurality of playback devices each having a generative media module therein; transmitting, via the network interface, timing data to the plurality of playback devices, such that the playback devices concurrently play back generative media content based at least in part on the input parameters.

**[0193]** Example 9: The device of any one of the Examples herein, wherein first and second playback devices play back different generative audio content, each based at least in part on the input parameters.

**[0194]** Example 10: The device of any one of the Examples herein, wherein the input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors, microphones); environmental data (e.g., weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of

users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity), user mood data).

**[0195]** Example 11: The device of any one of the Examples herein, wherein the timing data comprises at least one of: clock data or one or more synchronization signals.

**[0196]** Example 12: The device of any one of the Examples herein, wherein the operations further comprise transmitting, via the network interface, a signal from the coordinator device to at least one of the plurality of playback devices that causes the generative media module of the playback device to be modified.

**[0197]** Example 13: The device of any one of the Examples herein, wherein the generative media content comprises at least one of: generative audio content or generative visual content.

**[0198]** Example 14: The device of any one of the Examples herein, wherein the generative media modules comprise algorithms that automatically generate novel media output based on inputs that include at least the input parameters.

**[0199]** Example 15: A tangible, non-transitory computer-readable medium storing instructions that, when executed by one or more processors of a device cause the device to perform operations comprising: receiving, at a coordinator device, input parameters; transmitting the input parameters from the coordinator device to a plurality of playback devices each having a generative media module therein; transmitting timing data, from the coordinator device to the plurality of playback devices, such that the playback devices concurrently play back generative media content based at least in part on the input parameters.

**[0200]** Example 16: The computer-readable medium of any one of the Examples herein, wherein first and second playback devices play back different generative audio content, each based at least in part on the input parameters.

**[0201]** Example 17: The computer-readable medium of any one of the Examples herein, wherein the input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors, microphones); environmental data (e.g., weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity), user mood data).

**[0202]** Example 18: The computer-readable medium of any one of the Examples herein, wherein the timing data comprises at least one of: clock data or one or more synchronization signals.

**[0203]** Example 19: The computer-readable medium of any one of the Examples herein, further comprising transmitting a signal from the coordinator device to at least one of the plurality of playback devices that causes the generative media module of the playback device to be modified.

**[0204]** Example 20: The computer-readable medium of any one of the Examples herein, wherein the generative

media content comprises at least one of: generative audio content or generative visual content.

**[0205]** Example 21: The computer-readable medium of any one of the Examples herein, wherein the generative media modules comprise algorithms that automatically generate novel media output based on inputs that include at least the input parameters.

**[0206]** Example 22: A method comprising: receiving, at a coordinator device, input parameters; generating, via a generative media module of the coordinator device, first and second media content streams; transmitting, via the coordinator device, the first media content stream to a first playback device; transmitting, via the coordinator device, the second media content stream to a second playback device such that the first and second media content streams are played back concurrently via the first and second playback devices.

**[0207]** Example 23: The method of any one of the Examples herein, further comprising transmitting timing data from the coordinator device to each of the first and second playback devices.

**[0208]** Example 24: The method of any one of the Examples herein, wherein the timing data comprises at least one of: clock data or one or more synchronization signals.

**[0209]** Example 25: The method of any one of the Examples herein, wherein the first and second media content streams differ.

**[0210]** Example 26: The method of any one of the Examples herein, wherein the input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors); environmental data (e.g., weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity), user mood data).

**[0211]** Example 27: The method of any one of the Examples herein, further comprising modifying the generative media module of the coordinator device.

**[0212]** Example 28: The method of any one of the Examples herein, wherein each of the first and second generative media content streams comprises at least one of: generative audio content or generative visual content.

**[0213]** Example 29: The method of any one of the Examples herein, wherein the generative media modules comprises an algorithm that automatically generates novel media output based on inputs that include at least the input parameters.

**[0214]** Example 30: A device comprising: a network interface; a generative media module; one or more processors; and tangible, non-transitory computer-readable media storing instructions that, when executed by the one or more processors, cause the device to perform operations comprising: receiving, via the network interface, input parameters; generating, via the generative media module, first and second media content streams; transmitting, via the network interface, the first media content stream to a first playback

device; and transmitting, via the network interface, the second media content stream to a second playback device such that the first and second media content streams are played back concurrently via the first and second playback devices.

**[0215]** Example 31: The device of any one of the Examples herein, wherein the operations further comprise transmitting, via the network interface, timing data to each of the first and second playback devices.

**[0216]** Example 32: The device of any one of the Examples herein, wherein the timing data comprises at least one of: clock data or one or more synchronization signals.

**[0217]** Example 33: The device of any one of the Examples herein, wherein the first and second media content streams differ.

**[0218]** Example 34: The device of any one of the Examples herein, wherein the input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors); environmental data (e.g., weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity), user mood data).

**[0219]** Example 35: The device of any one of the Examples herein, wherein the operations further comprise modifying the generative media module.

**[0220]** Example 36: The device of any one of the Examples herein, wherein each of the first and second generative media content streams comprises at least one of: generative audio content or generative visual content.

**[0221]** Example 37: The device of any one of the Examples herein, wherein the generative media modules comprises an algorithm that automatically generates novel media output based on inputs that include at least the input parameters.

**[0222]** Example 38: A tangible, non-transitory computer-readable medium storing instructions that, when executed by one or more processors of a coordinator device, cause the coordinator device to perform operations comprising: receiving, at the coordinator device, input parameters; generating, via a generative media module of the coordinator device, first and second media content streams; transmitting, via the coordinator device, the first media content stream to a first playback device; and transmitting, via the coordinator device, the second media content stream to a second playback device such that the first and second media content streams are played back concurrently via the first and second playback devices.

**[0223]** Example 39: The computer-readable medium of any one of the Examples herein, further comprising transmitting timing data from the coordinator device to each of the first and second playback devices.

**[0224]** Example 40: The computer-readable medium of any one of the Examples herein, wherein the timing data comprises at least one of: clock data or one or more synchronization signals.

**[0225]** Example 41: The computer-readable medium of any one of the Examples herein, wherein the first and second media content streams differ.

**[0226]** Example 42: The computer-readable medium of any one of the Examples herein, wherein the input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors); environmental data (e.g., weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity), user mood data).

**[0227]** Example 43: The computer-readable medium of any one of the Examples herein, wherein the operations further comprise modifying the generative media module of the coordinator device.

**[0228]** Example 44: The computer-readable medium of any one of the Examples herein, wherein each of the first and second generative media content streams comprises at least one of: generative audio content or generative visual content.

**[0229]** Example 45: The computer-readable medium of any one of the Examples herein, wherein the generative media modules comprises an algorithm that automatically generates novel media output based on inputs that include at least the input parameters.

**[0230]** Example 46: A playback device comprising: one or more amplifiers configured to drive one or more audio transducers; one or more processors; and data storage having instructions thereon that, when executed by the one or more processors, cause the playback device to perform operations comprising: receiving, at the playback device, one or more first input parameters; generating, via the playback device, first media content based at least in part on the one or more first input parameters, the generating comprising: accessing a library stored on the playback device including a plurality of pre-existing media segments; and arranging a first selection of pre-existing media segments from the library for playback according to a generative media content model and based at least in part on the one or more input parameters; and playing back, via the one or more amplifiers, the first generated media content.

**[0231]** Example 47: The playback device of any one of the Examples herein, wherein the operations further comprise: receiving, at the playback device, one or more second input parameters different from the first; generating, via the playback device, second media content based at least in part on the one or more second input parameters, the second media content different from the first, the generating comprising: accessing the library; and arranging a second selection of pre-existing media segments from the library for playback according to the generative media content model and based at least in part on the one or more second input parameters; and playing back, via the one or more amplifiers, the second generated media content.

**[0232]** Example 48: The playback device of claim 1, wherein arranging the first selection of pre-existing media

segments from the library for playback comprises arranging two or more of the pre-existing media segments in an at least partially temporally offset manner.

**[0233]** Example 49: The playback device of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in at least partially temporally overlapping manner.

**[0234]** Example 50: The playback device of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library for playback comprises applying different equalization adjustments to different pre-existing media segments.

**[0235]** Example 51: The playback device of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library or playback comprises applying varying gain levels over time to different pre-existing media segments.

**[0236]** Example 52: The playback device of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library or playback comprises randomizing a start point for playback of a particular pre-existing media segment.

**[0237]** Example 53: The playback device of any one of the Examples herein, wherein the first generated media content and the second generated media content each comprises novel media content.

**[0238]** Example 54: The playback device of any one of the Examples herein, wherein the first generated media content comprises audio content and the plurality of pre-existing media segments comprises a plurality of pre-existing audio segments.

**[0239]** Example 55: The playback device of any one of the Examples herein, wherein the first generated media content comprises audio-visual content and the plurality of pre-existing media segments comprises a plurality of pre-existing audio segments, pre-existing visual media segments, or pre-existing audio-visual media segments.

**[0240]** Example 56: The playback device of any one of the Examples herein, further comprising: receiving, via a network interface, additional pre-existing media segments; and updating the library to include at least the additional pre-existing media segments.

**[0241]** Example 57: The playback device of any one of the Examples herein, wherein the first and second input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors, microphones); environmental (e.g., data weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity, voice utterance characteristics), user mood data).

**[0242]** Example 58: A method comprising: receiving, at a playback device, one or more first input parameters; generating, via the playback device, first media content based at

least in part on the one or more first input parameters, the generating comprising: accessing a library stored on the playback device including a plurality of pre-existing media segments; and arranging a first selection of pre-existing media segments from the library for playback according to a generative media content model and based at least in part on the one or more input parameters; and playing back, via the playback device, the first generated media content.

[0243] Example 59: The method of any one of the Examples herein, further comprising: receiving, at the playback device, one or more second input parameters different from the first; generating, via the playback device, second media content based at least in part on the one or more second input parameters, the second media content different from the first, the generating comprising: accessing the library; and arranging a second selection of pre-existing media segments from the library for playback according to the generative media content model and based at least in part on the one or more second input parameters; and playing back, via the playback device, the second generated media content.

[0244] Example 60: The method of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in an at least partially temporally offset manner.

[0245] Example 61: The method of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in at least partially temporally overlapping manner.

[0246] Example 62: The method of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library for playback comprises applying different equalization adjustments to different pre-existing media segments.

[0247] Example 63: The method of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library or playback comprises applying varying gain levels over time to different pre-existing media segments.

[0248] Example 64: The method of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library or playback comprises randomizing a start point for playback of a particular pre-existing media segment.

[0249] Example 65: The method of any one of the Examples herein, wherein the first generated media content and the second generated media content each comprises novel media content.

[0250] Example 66: The method of any one of the Examples herein, wherein the first generated media content comprises audio content and the plurality of pre-existing media segments comprises a plurality of pre-existing audio segments.

[0251] Example 67: The method of any one of the Examples herein, wherein the first generated media content comprises audio-visual content and the plurality of pre-existing media segments comprises a plurality of pre-existing audio segments, pre-existing visual media segments, or pre-existing audio-visual media segments.

[0252] Example 68: The method of any one of the Examples herein, further comprising: receiving, via a net-

work interface, additional pre-existing media segments; and updating the library to include at least the additional pre-existing media segments.

[0253] Example 69: The method of any one of the Examples herein, wherein the first and second input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors, microphones); environmental data (e.g., weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity, voice utterance characteristics), user mood data).

[0254] Example 70: A tangible, non-transitory, computer-readable media storing instructions that, when executed by one or more processors of a playback device, cause the playback device to perform operations comprising: receiving, at the playback device, one or more first input parameters; generating, via the playback device, first media content based at least in part on the one or more first input parameters, the generating comprising: accessing a library stored on the playback device including a plurality of pre-existing media segments; and arranging a first selection of pre-existing media segments from the library for playback according to a generative media content model and based at least in part on the one or more input parameters; and playing back, via the playback device, the first generated media content.

[0255] Example 71: The computer-readable media of any one of the Examples herein, wherein the operations further comprise: receiving, at the playback device, one or more second input parameters different from the first; generating, via the playback device, second media content based at least in part on the one or more second input parameters, the second media content different from the first, the generating comprising: accessing the library; and arranging a second selection of pre-existing media segments from the library for playback according to the generative media content model and based at least in part on the one or more second input parameters; and playing back, via the one or more amplifiers, the second generated media content.

[0256] Example 72: The computer-readable media of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in an at least partially temporally offset manner.

[0257] Example 73: The computer-readable media of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in at least partially temporally overlapping manner.

[0258] Example 74: The computer-readable media of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library for

playback comprises applying different equalization adjustments to different pre-existing media segments.

**[0259]** Example 75: The computer-readable media of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library or playback comprises applying varying gain levels over time to different pre-existing media segments.

**[0260]** Example 76: The computer-readable media of any one of the Examples herein, wherein arranging the first selection of pre-existing media segments from the library or playback comprises randomizing a start point for playback of a particular pre-existing media segment.

**[0261]** Example 77: The computer-readable media of any one of the Examples herein, wherein the first generated media content and the second generated media content each comprises novel media content.

**[0262]** Example 78: The computer-readable media of any one of the Examples herein, wherein the first generated media content comprises audio content and the plurality of pre-existing media segments comprises a plurality of pre-existing audio segments.

**[0263]** Example 79: The computer-readable media of any one of the Examples herein, wherein the first generated media content comprises audio-visual content and the plurality of pre-existing media segments comprises a plurality of pre-existing audio segments, pre-existing visual media segments, or pre-existing audio-visual media segments.

**[0264]** Example 80: The computer-readable media of any one of the Examples herein, further comprising: receiving, via a network interface, additional pre-existing media segments; and updating the library to include at least the additional pre-existing media segments.

**[0265]** Example 81: The computer-readable media of any one of the Examples herein, wherein the first and second input parameters comprise one or more of: physiological sensor data (e.g., biometric sensors, wearable sensors (heart rate, temperature, breathing rate, brainwave)); networked device sensor data (e.g., cameras, lights, temperature sensors, thermostats, presence detectors, microphones); environmental data (e.g., weather, temperature, time/day/week/month); playback device capability data (e.g., number and type of transducers, output power); playback device state (e.g., device temperature, battery level, current audio playback, playback device location, whether playback device is bonded with another playback device); or user data (e.g., user identity, number of users present, user location, user history data, user preference data, user biometric data (heart rate, temperature, breathing rate, brain activity, voice utterance characteristics), user mood data).

**[0266]** Example 82: A system, comprising a first playback device and a second playback device. The first playback device comprises: a first network interface; one or more first processors; and data storage having instructions thereon that, when executed by the one or more processors, cause the first playback device to perform operations comprising: receiving one or more input parameters; generating media content based at least in part on the one or more input parameters, the generated media content comprising a first portion and at least a second portion, the generating comprising: accessing a library stored on the playback device including a plurality of pre-existing media segments; and arranging a selection of pre-existing media segments from the library for playback according to a generative media content model and based at least in part on the one or more

input parameters; transmitting, via the first network interface, a signal comprising the second portion of the generated media content and corresponding timing information; and causing playback of the first portion of the generated media content. The second playback device comprises: a second network interface; one or more audio transducers; one or more second processors; and data storage having instructions thereon that, when executed by the one or more second processors, cause the second playback device to perform operations comprising: receiving, via the second network interface, the transmitted signal from the first playback device; and playing back, via the one or more transducers, the second portion of the generated media content according to the timing information in substantial synchrony with playback of the first portion of the generated media content.

**[0267]** Example 83: The system of any one of the Examples herein, further comprising: a network device, comprising: a third network interface; one or more processors; and data storage having instructions thereon that, when executed by the one or more processors, cause the third playback device to perform operations comprising: receiving, via the third network interface a data network, a request from the first playback device; and in response to receiving the request, transmitting, via the third network interface over the data network, an updated library of pre-existing media segments to the first playback device.

**[0268]** Example 84: The system of any one of the Examples herein, wherein the network device comprises one or more of: a remote server, another playback device, a mobile computing device, a laptop, or a tablet.

**[0269]** Example 85: A system comprises a first playback device a second playback device communicatively coupled over a local area network. The first playback device comprises: one or more first processors; one or more first audio transducers; and data storage having instructions thereon that, when executed by the one or more first processors, cause the first playback device to perform operations comprising: receiving one or more input parameters; generating first media content based at least in part on the one or more input parameters, the generating comprising: accessing a first library stored on the first playback device including a plurality of pre-existing media segments; and arranging a selection of pre-existing media segments from the first library for playback according to a first generative media content model and based at least in part on the one or more input parameters; and playing back, via the one or more first audio transducers, the first generative media content. The second playback device comprises: a second network interface;

**[0270]** one or more second audio transducers; one or more second processors; and data storage having instructions thereon that, when executed by the one or more second processors, cause the second playback device to perform operations comprising: generating second media content based at least in part on the one or more input parameters, the second generated media content being substantially identical to the first generated media content, the generating comprising: accessing a second library stored on the second playback device including a plurality of pre-existing media segments; and arranging a selection of pre-existing media segments from the second library for playback according to a second generative media content model and based at least in part on the one or more input param-

eters; and playing back, via the one or more second audio transducers, the second generated media content in synchrony with playback of the first generated media content via the first playback device.

**[0271]** Example 86: The system of any one of the Examples herein, wherein the first generative media content model and the second generative media content model are substantially identical.

**[0272]** Example 87: The system of any one of the Examples herein, wherein the first library and the second library are substantially identical.

**[0273]** Example 88: A method comprising: accessing, via a playback device, blockchain data stored on a distributed ledger; generating, via the playback device, media content based at least in part on the blockchain data, the generating comprising: accessing a library stored on the playback device including a plurality of pre-existing media segments; and arranging a selection of pre-existing media segments from the library for playback according to a generative media content model and based at least in part on the blockchain data; and playing back, via the playback device, the generated media content.

**[0274]** Example 89: The method of any one of the preceding Examples, wherein the NFT data comprises one or more pre-existing media segments, and wherein accessing the NFT data comprises storing the one or more pre-existing media segments in the library.

**[0275]** Example 90: The method of any one of the preceding Examples, wherein the blockchain data comprises first non-fungible token (NFT) data, wherein the distributed ledger is a first distributed ledger, the method further comprising accessing, via the playback device, data associated with a second NFT stored on a second distributed ledger, and wherein arranging the selection of pre-existing media segments from the library for playback according to the generative media content model is based at least in part on both the first NFT data and the second NFT data.

**[0276]** Example 91: The method of any one of the preceding Examples, wherein the first distributed ledger is associated with a first blockchain layer, and wherein the second distributed ledger is associated with a second blockchain layer different from the first blockchain layer.

**[0277]** Example 92: The method of any one of the preceding Examples, wherein the blockchain data is associated with a playlist.

**[0278]** Example 93: The method of any one of the preceding Examples, wherein the blockchain data depends at least in part on transactions recorded on the distributed ledger involving a non-fungible token (NFT).

**[0279]** Example 94: The method of any one of the preceding Examples, wherein arranging the selection of pre-existing media segments from the library according to the generative media content model is further based at least in part on one or more input parameters.

**[0280]** Example 95: The method of any one of the preceding Examples, wherein the input parameters comprise one or more of: physiological sensor data; networked device sensor data; environmental data; playback device characteristic data; playback device state; user listening history data; oracle data stored via a distributed ledger, or user data.

**[0281]** Example 96: The method of any one of the preceding Examples, wherein user listening history data is stored via a distributed ledger.

**[0282]** Example 97: The method of any one of the preceding Examples, wherein accessing the blockchain data comprises connecting to a user wallet that holds a non-fungible token (NFT).

**[0283]** Example 98: The method of any one of the preceding Examples, wherein accessing the blockchain data comprises accessing a code associated with a physical media object via a control device (e.g., QR code or other code imprinted on custom vinyl or other media).

**[0284]** Example 99: The method of any one of the preceding Examples, wherein arranging the selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in an at least partially temporally offset manner.

**[0285]** Example 100: The method of any one of the preceding Examples, wherein arranging the selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in at least partially temporally overlapping manner.

**[0286]** Example 101: A method comprising: transmitting, via a playback device, data associated with a first token over a network to a network address of a distributed ledger, the address associated with a generative media smart contract configured to produce a generative media content model; receiving, via the playback device, the generative media content model from the network address associated with the generative media smart contract; generating, via the playback device, media content based at least in part on the generative media content model, the generating comprising: accessing a library comprising a plurality of pre-existing media segments; and arranging a selection of pre-existing media segments from the library for playback according to the generative media content model; and playing back, via the playback device, the generated media content.

**[0287]** Example 102: The method of any one of the preceding Examples wherein the token data comprises first non-fungible token (NFT) data, the method further comprising: transmitting, via the playback device, data associated with a second NFT stored on a distributed ledger to the network address associated with the generative media smart contract; receiving, via the playback device, a second generative media content model from the network address associated with the generative media smart contract, the second generative media content model being different from the first; generating, via the playback device, second media content based at least in part on the second generative media content model; and playing back, via the playback device, the second generated media content.

**[0288]** Example 103: The method of any one of the preceding Examples, wherein the token data is associated with a curated playlist.

**[0289]** Example 104: The method of any one of the preceding Examples, wherein the token data depends at least in part on transactions recorded on the distributed ledger involving the token.

**[0290]** Example 105: The method of any one of the preceding Examples, wherein arranging the selection of pre-existing media segments from the library according to the generative media content model is further based at least in part on one or more input parameters.

**[0291]** Example 106: The method of any one of the preceding Examples, wherein the input parameters comprise one or more of: physiological sensor data; networked device

sensor data; environmental data; playback device characteristic data; playback device state; user listening history data; oracle data stored via a distributed ledger, or user data.

[0292] Example 107: The method of any one of the preceding Examples, wherein user listening history data is stored via a distributed ledger.

[0293] Example 108: The method of any one of the preceding Examples, further comprising, before transmitting the token data, accessing the token data by connecting to a user wallet that stores the token data.

[0294] Example 109: The method of any one of the preceding Examples, further comprising, before transmitting the token data, accessing the token data via a code associated with a physical media object via a control device (e.g., QR code or other code imprinted on custom vinyl or other media)

[0295] Example 110: The method of any one of the preceding Examples wherein arranging the selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in an at least partially temporally offset manner.

[0296] Example 111: The method of any one of the preceding Examples, wherein arranging the selection of pre-existing media segments from the library for playback comprises arranging two or more of the pre-existing media segments in at least partially temporally overlapping manner.

[0297] Example 112: The method of any one of the preceding Examples, wherein the first generated media content and the second generated media content each comprises novel media content.

[0298] Example 113: One or more tangible, non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause a media playback system or a playback device to perform operations comprising: the method of any one of the preceding Examples.

[0299] Example 114: A media playback system comprising: one or more processors; and the computer-readable media of any one of the preceding Examples.

[0300] Example 115: A playback device comprising: one or more processors; and the computer-readable media of any one of the preceding Examples.

1. (canceled)

2. A computing system, comprising:

a network interface;

one or more processors; and

memory storing instructions that, when executed by the one or more processors, cause the computing system to perform operations comprising:

receiving, via the network interface, first data via a distributed ledger, wherein the first data corresponds to one or more digital tokens associated with a real-world asset;

generating, via a generative media module, an output based on the received first data; and

causing, via the network interface, one or more network devices to perform an operation corresponding to the generated output.

3. The computing system of claim 2, wherein the first data comprises a transaction history associated with the real-world asset.

4. The computing system of claim 2, wherein the first data comprises third-party data associated with the real-world asset.

5. The computing system of claim 4, wherein the third-party data comprises at least one of a price or other value associated with the real-world asset.

6. The computing system of claim 4, wherein the third-party data is received via an oracle.

7. The computing system of claim 2, wherein the real-world asset comprises a tangible asset.

8. The computing system of claim 7, wherein the tangible asset comprises property.

9. The computing system of claim 2, wherein the real-world asset comprises an intangible asset.

10. The computing system of claim 9, wherein the intangible asset comprises currency.

11. The computing system of claim 2, wherein receiving the first data comprises receiving the first data via an address associated with a smart contract.

12. The computing system of claim 11, wherein the smart contract is associated with an entity that owns the real-world asset, and wherein the smart contract indicates proportional ownership interests in the real-world asset.

13. The computing system of claim 2, wherein the one or more digital tokens are stored in a digital wallet.

14. The computing system of claim 2, wherein the generated output comprises preexisting media content.

15. The computing system of claim 2, wherein the generated output comprises at least one of a playlist or a radio station.

16. The computing system of claim 2, wherein the generated output comprises synthetic media content.

17. The computing system of claim 16, wherein the generated synthetic media content comprises at least one of text, audio, music, spoken word, images, video, or tactile output.

18. The computing system of claim 16, wherein the generated synthetic media content comprises a light output having at least one of a particular hue, color temperature, brightness, or pattern.

19. The computing system of claim 2, wherein the operations further comprise sending, after generating the output, second data associated with the generated output to the address associated with the smart contract.

20. The computing system of claim 2, wherein causing the one or more network devices to perform the operation corresponding to the generated output comprises causing at least one of the one or more network devices to play back the synthetic media content.

21. A method comprising:

receiving, via a computing system, first data via a distributed ledger, wherein the first data corresponds to one or more digital tokens associated with a real-world asset;

generating, via a generative media module of the computing system, an output based on the received first data; and

causing, via the computing system, one or more network devices to perform an operation corresponding to the generated output.

22. One or more tangible, non-transitory computer-readable media storing instructions that, when executed by one or more processors of a computing system, cause the computing system to perform operations comprising:

receiving first data via a distributed ledger, wherein the first data corresponds to one or more digital tokens associated with a real-world asset;

generating, via a generative media module of the computing system, an output based on the received first data; and

causing one or more network devices to perform an operation corresponding to the generated output.

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